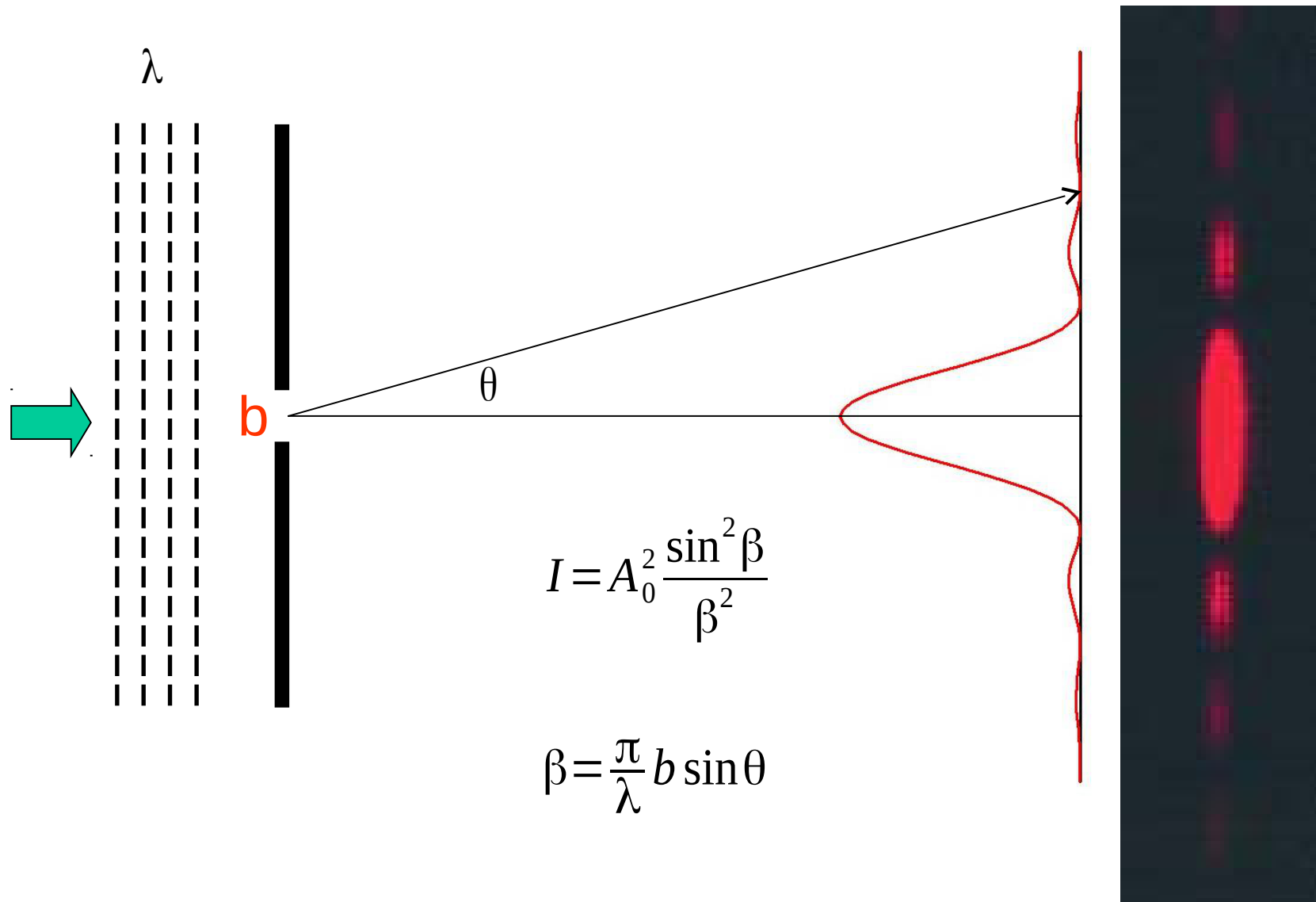


Introduction to Astronomical Spectroscopy

Stefano Ciroi

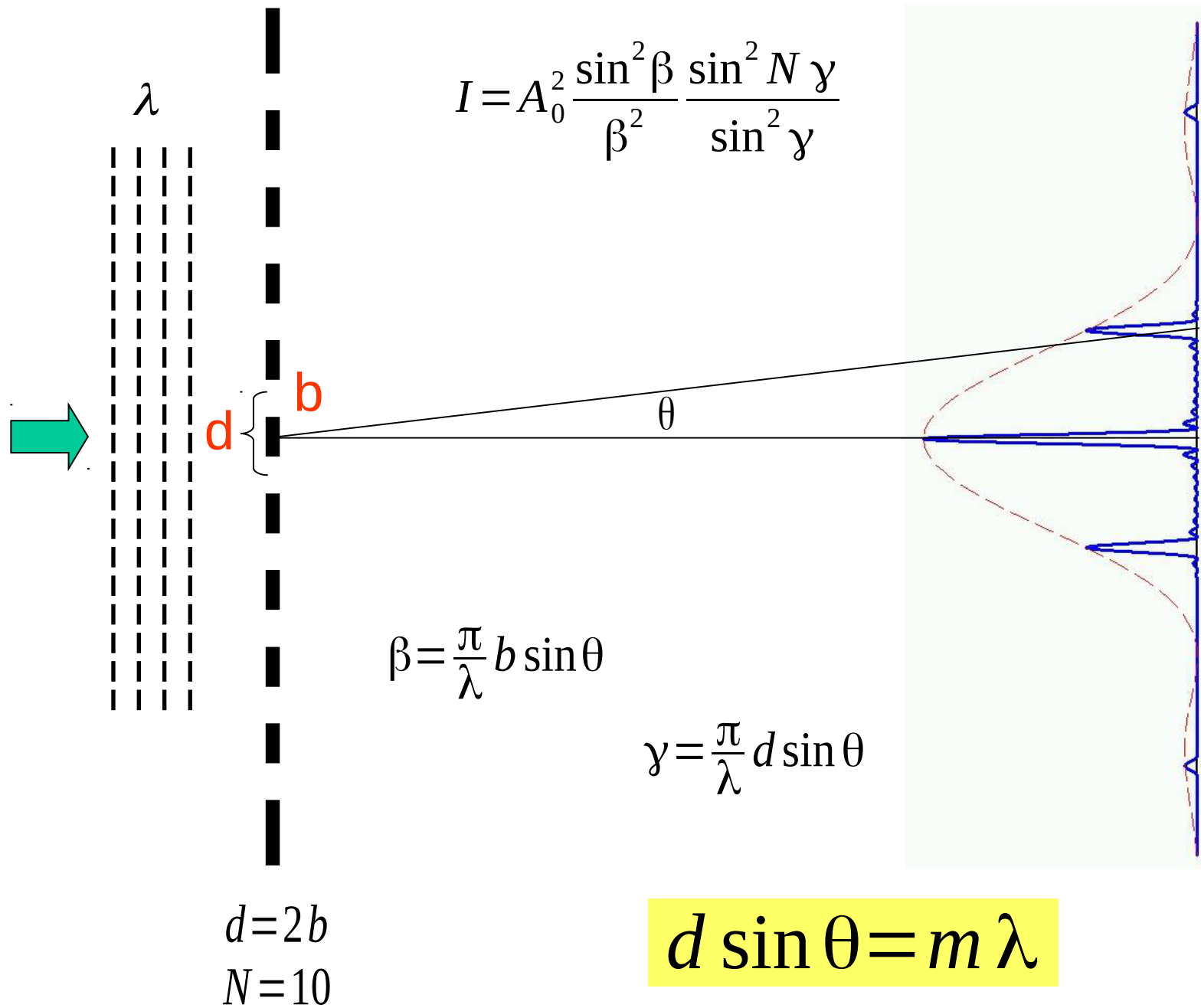
Dipartimento di Fisica e Astronomia
Università di Padova

Single slit diffraction



$$I=0 \Rightarrow \beta = \pm\pi, \pm2\pi, \dots, m\pi \Rightarrow b \sin \theta = m\lambda$$

N slits interference



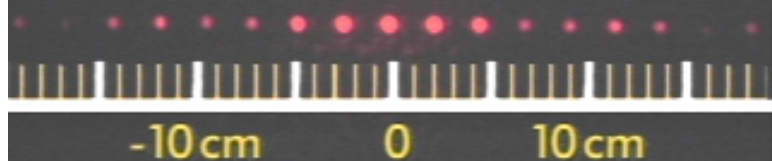
Slit spacing = .62 mm



Slit spacing = .09 mm



80 slits/cm



320 slits/cm



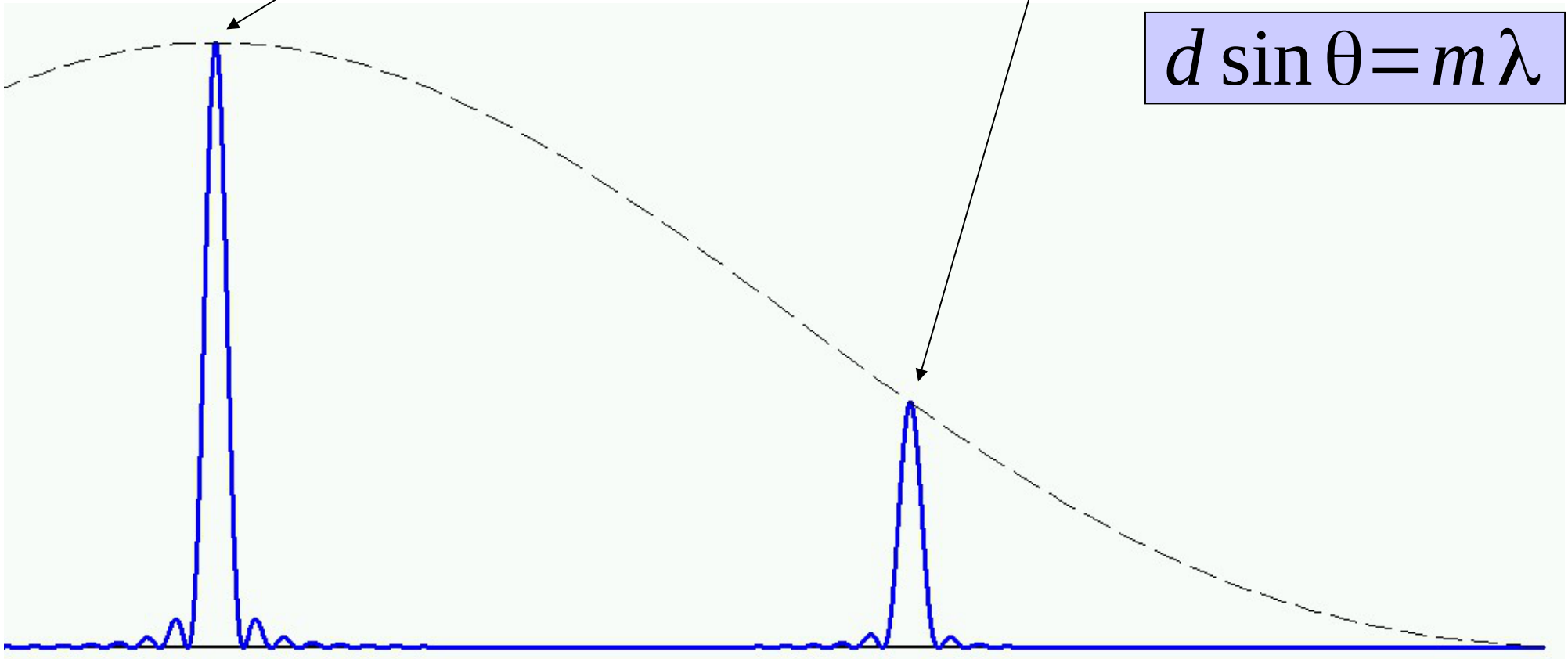
The spectrum

$$d = 3b$$
$$N = 25$$

Spectral line
 $\lambda = 4400 \text{ \AA}$
 $m = 0$

Spectral line
 $\lambda = 4400 \text{ \AA}$
 $m = 1$

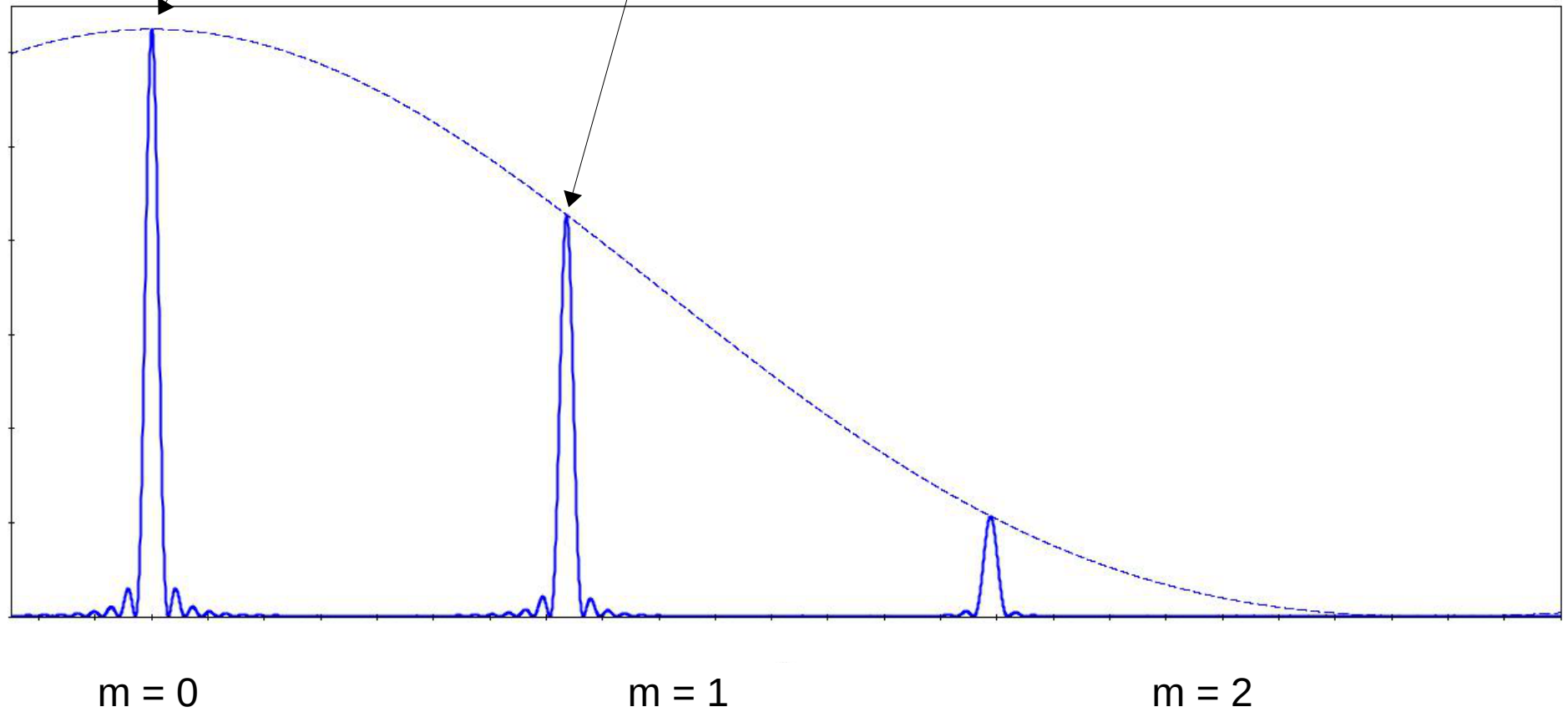
$$d \sin \theta = m \lambda$$



$$d \sin \theta = m \lambda$$

$\lambda = 4400 \text{ \AA}$

$\lambda = 4400 \text{ \AA}$

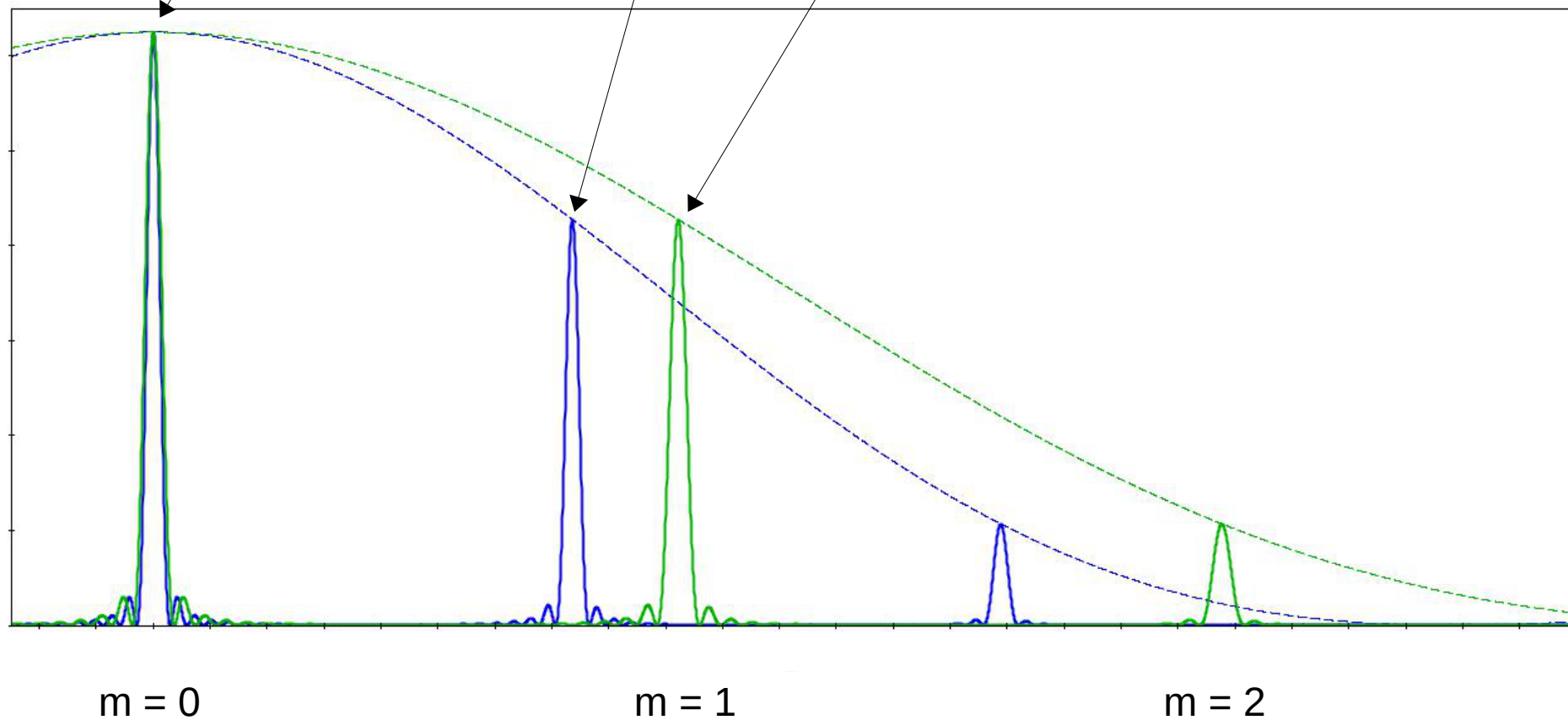


$$d \sin \theta = m \lambda$$

$\lambda = 4400 \text{ \AA} = 5500 \text{ \AA}$

$\lambda = 4400 \text{ \AA}$

$\lambda = 5500 \text{ \AA}$



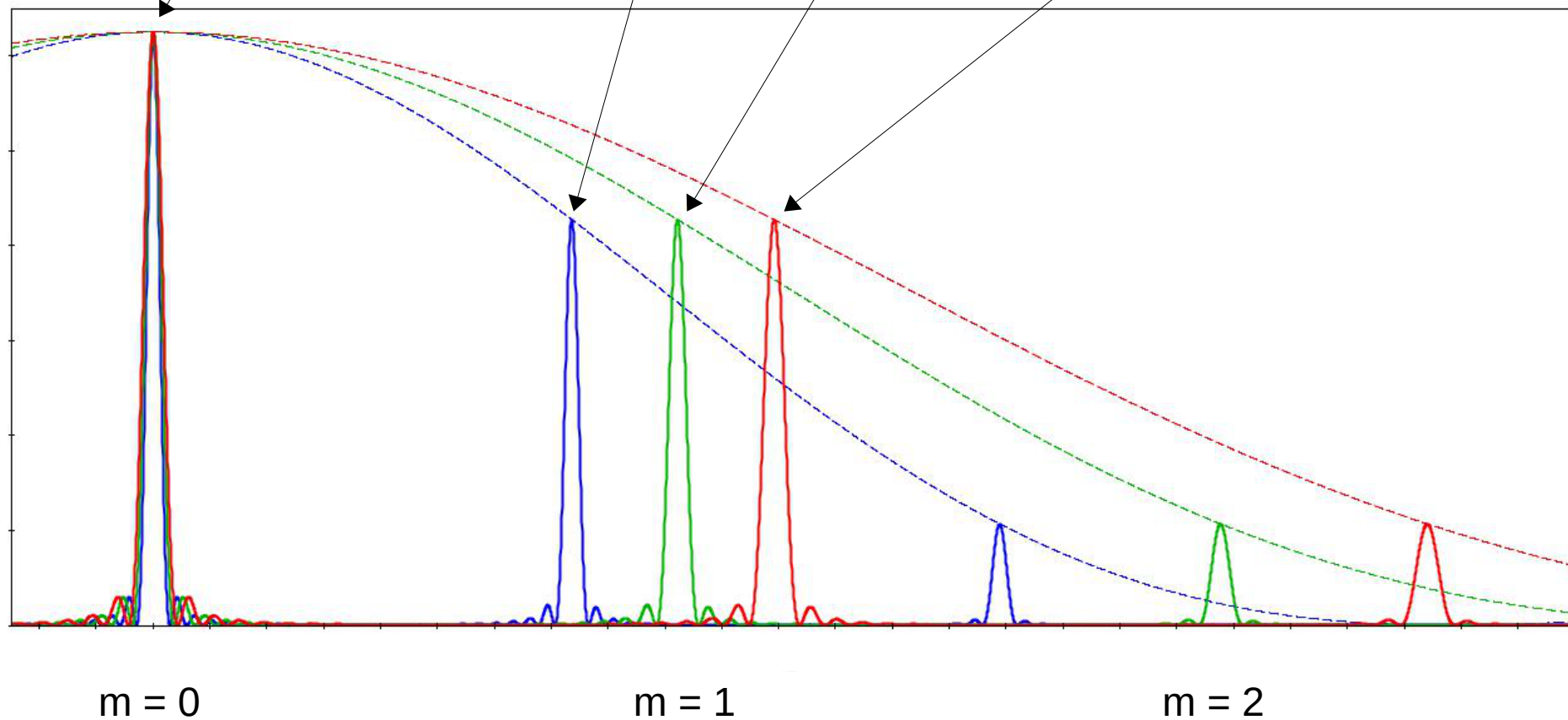
$$d \sin \theta = m \lambda$$

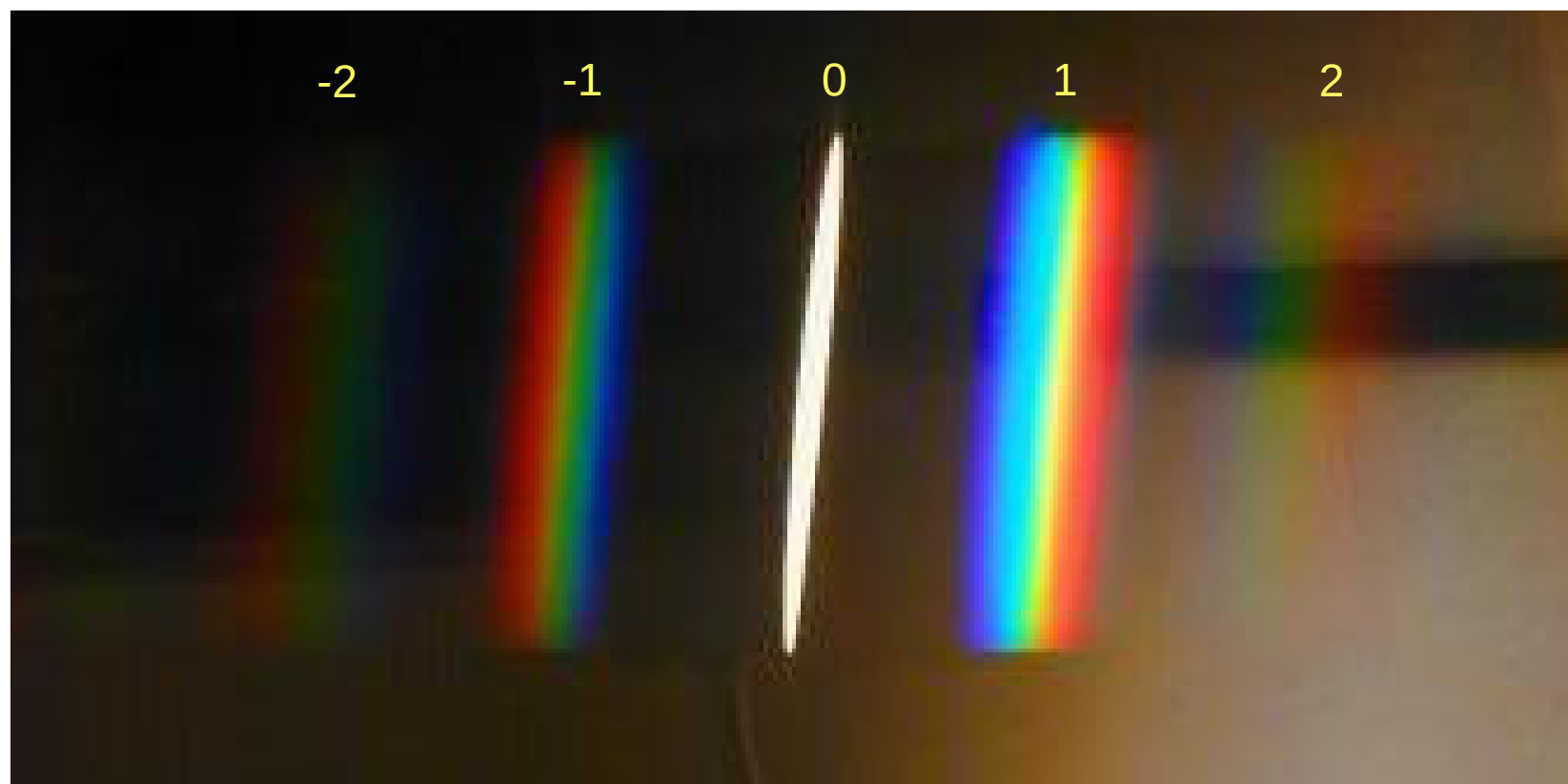
$$\lambda = 4400 \text{ \AA} = 5500 \text{ \AA} = 6500 \text{ \AA}$$

$$\lambda = 4400 \text{ \AA}$$

$$\lambda = 5500 \text{ \AA}$$

$$\lambda = 6500 \text{ \AA}$$



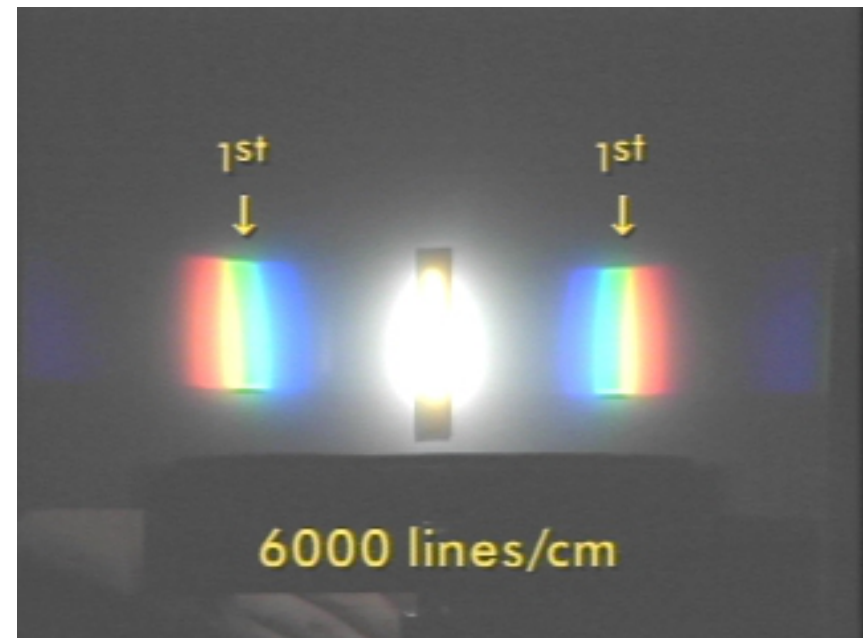
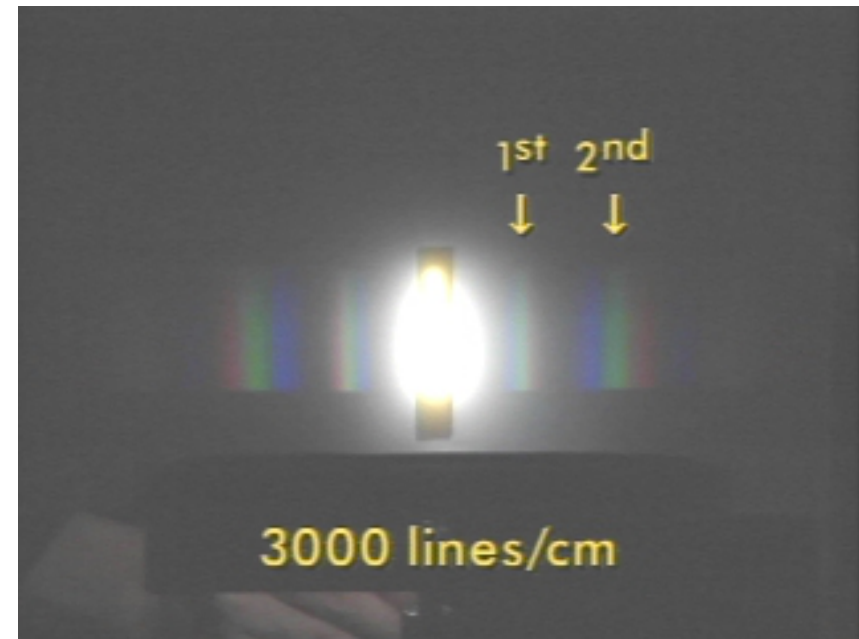


Grating equation

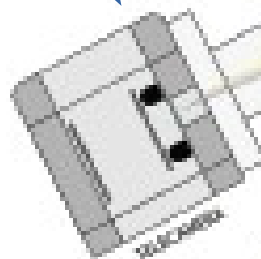
Inclination angle of the light beam

$$\sin \theta + \sin i = \rho m \lambda$$

Density of grating lines



guide
camera



slit

OTTURATORE

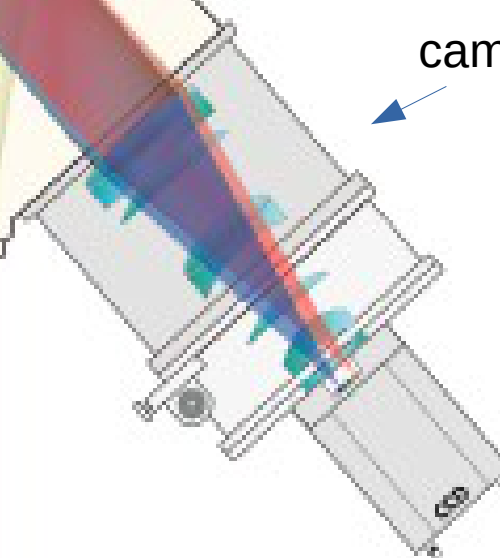
REMOVER

ARMATA PORTAFILTRI

RETICULO

grating

camera

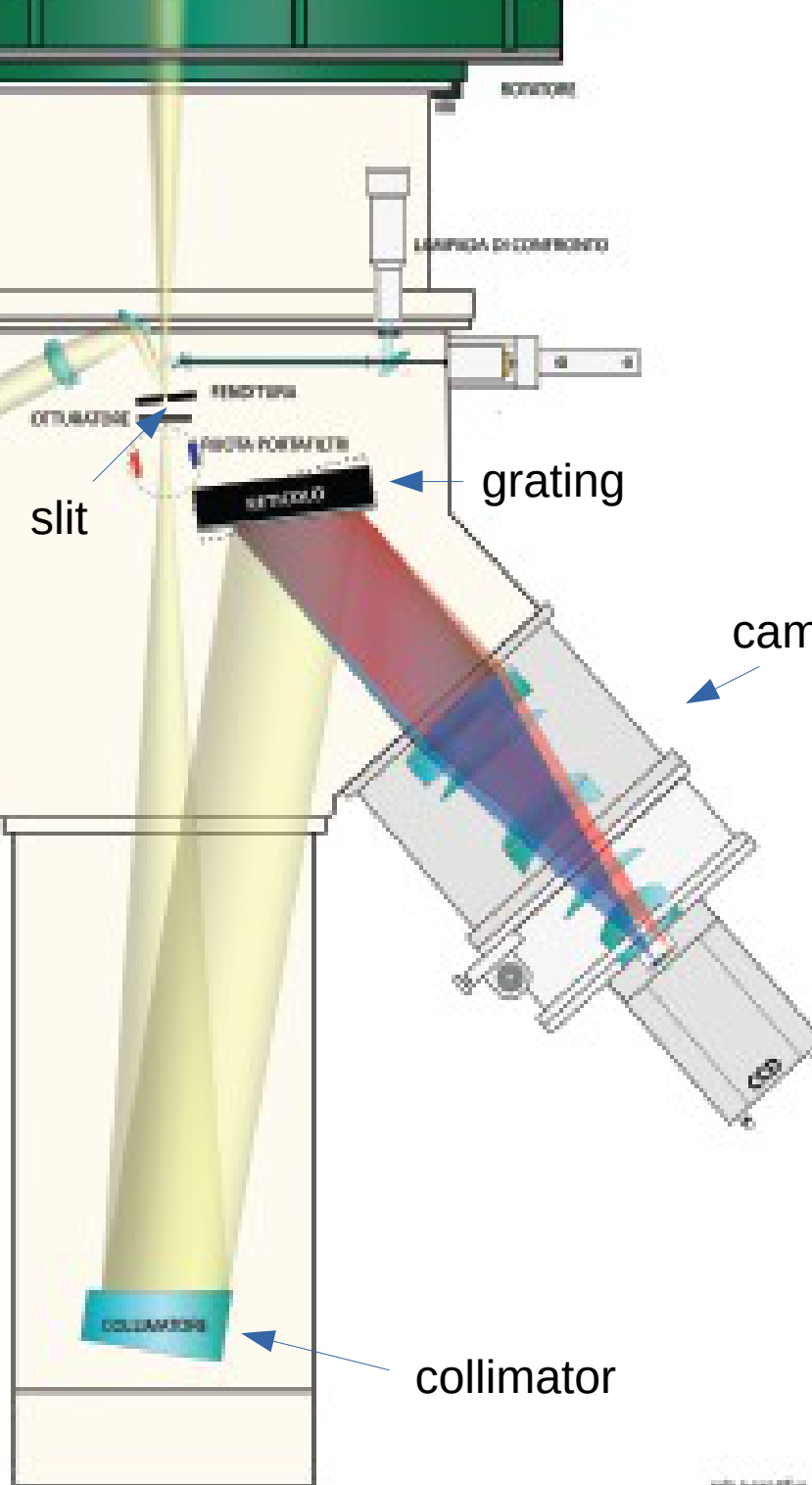


Boller & Chivens
Spectrograph

T120 - Asiago

COLLIMATORE

collimator



Dispersion and Resolution

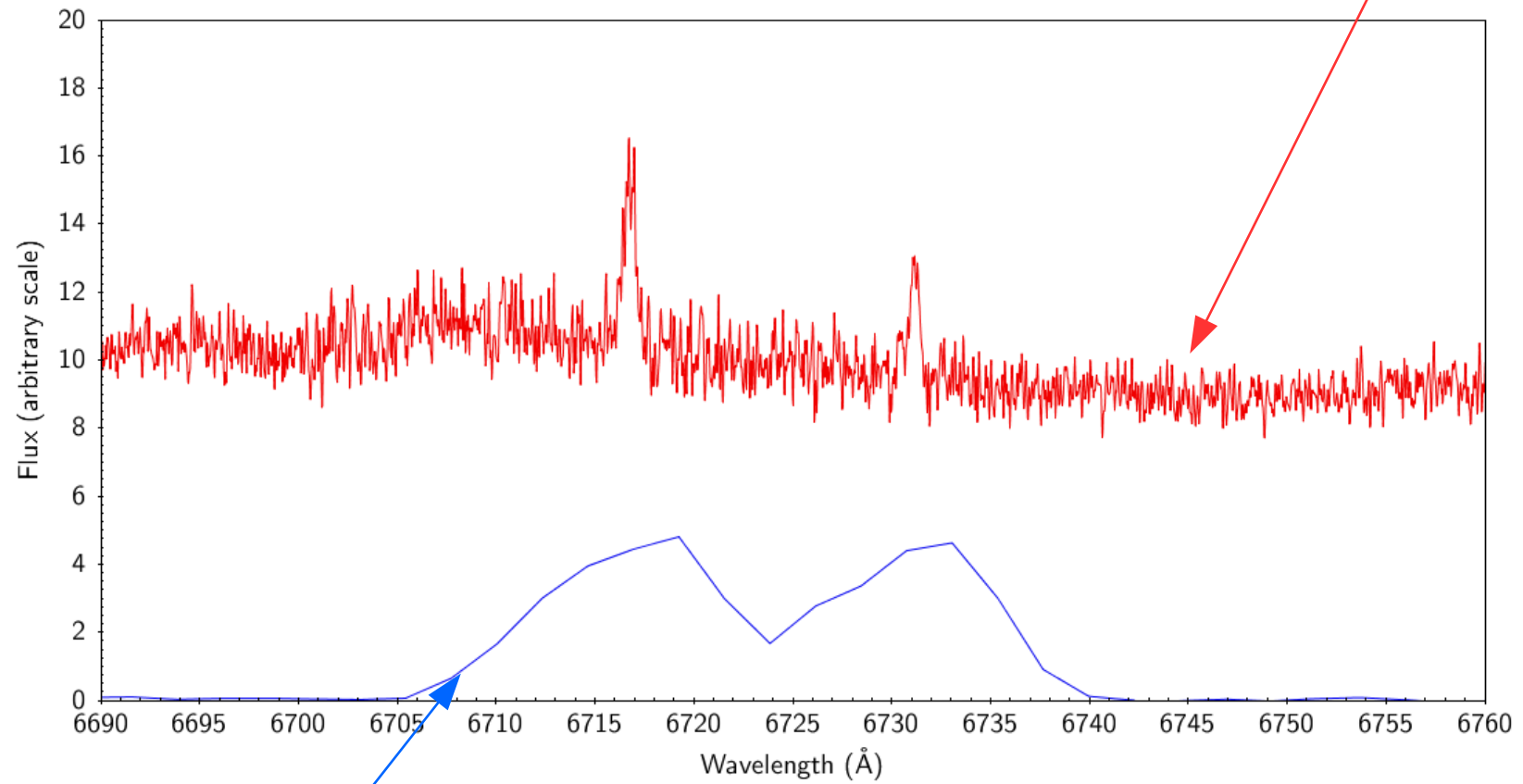
$$\frac{\Delta \lambda}{\Delta \theta} = \frac{\cos \theta}{\rho m} \qquad \frac{\Delta \lambda}{\Delta x} = \frac{\cos \theta}{\rho m f_{\text{cam}}}$$

$$R = \frac{\lambda}{\Delta \lambda} = \rho m W \frac{F_{\text{coll}} \lambda}{s} = \rho m W \frac{F_{\text{coll}} \lambda}{\alpha f_{\text{tel}}}$$

$$R = \frac{D_{\text{cam}}}{D_{\text{tel}}} \frac{\lambda}{\alpha \frac{\Delta \lambda}{\Delta \theta}}$$

$$\frac{\Delta\lambda}{\Delta x} = 0.03 \text{ \AA/px}$$

$$R = 15000$$

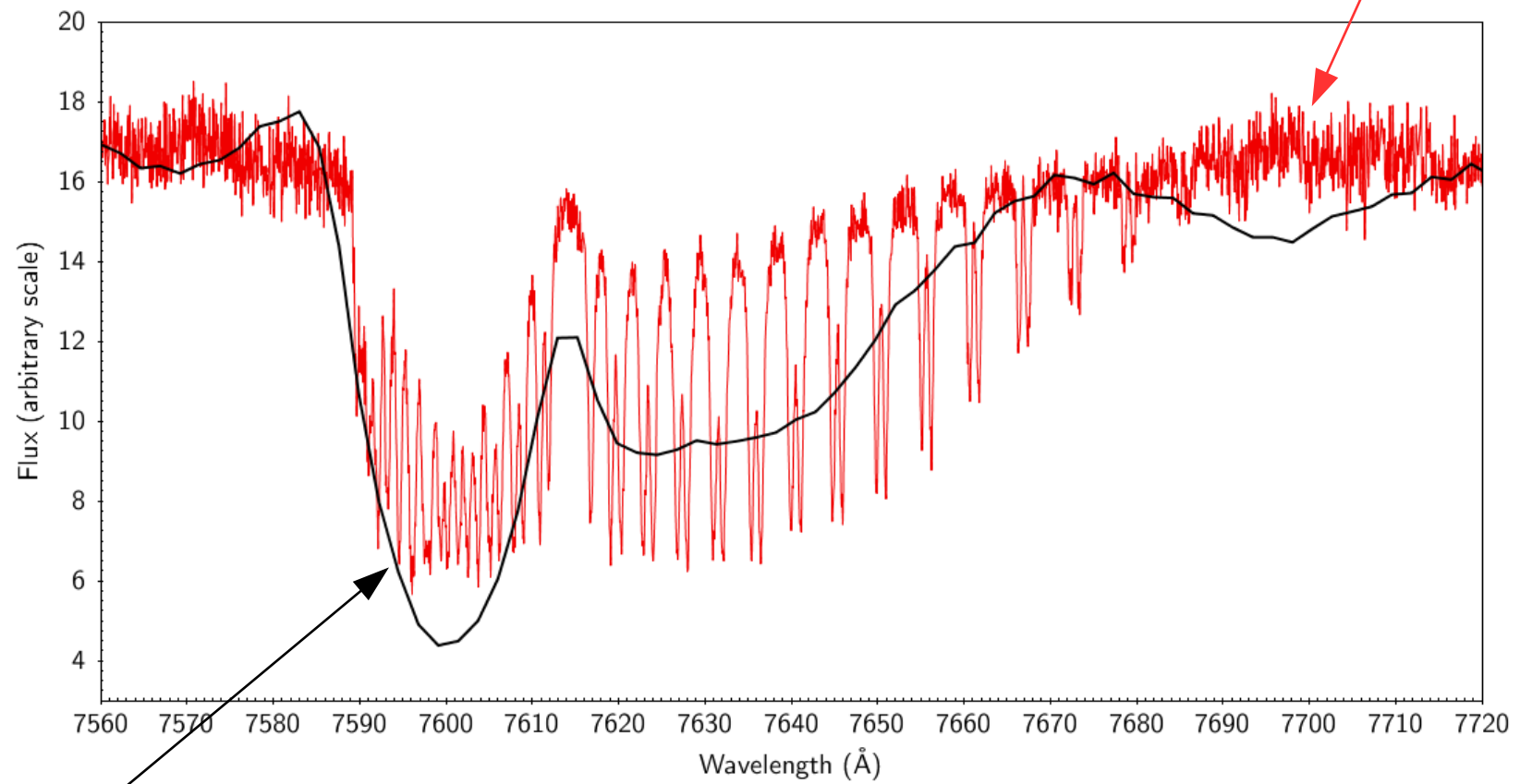


$$\frac{\Delta\lambda}{\Delta x} = 2.3 \text{ \AA/px}$$

$$R = 600$$

$$\frac{\Delta\lambda}{\Delta x} = 0.03 \text{ \AA/px}$$

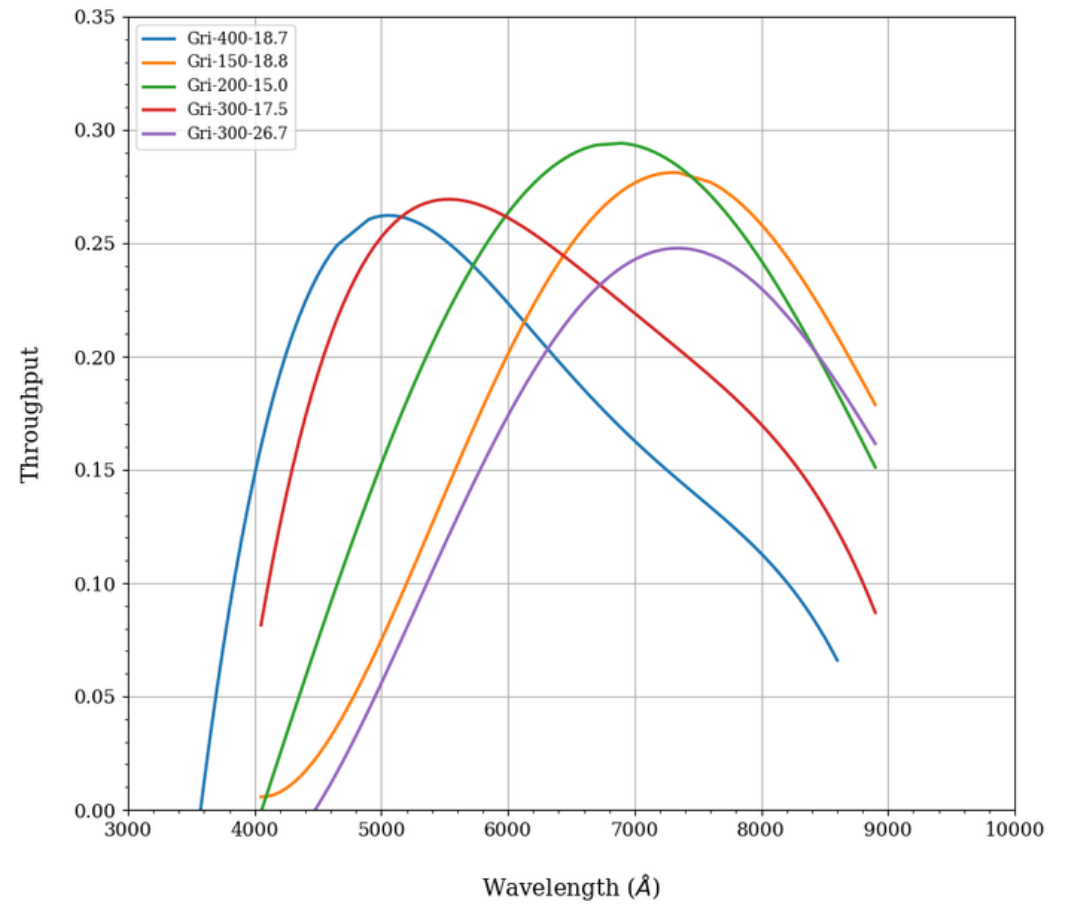
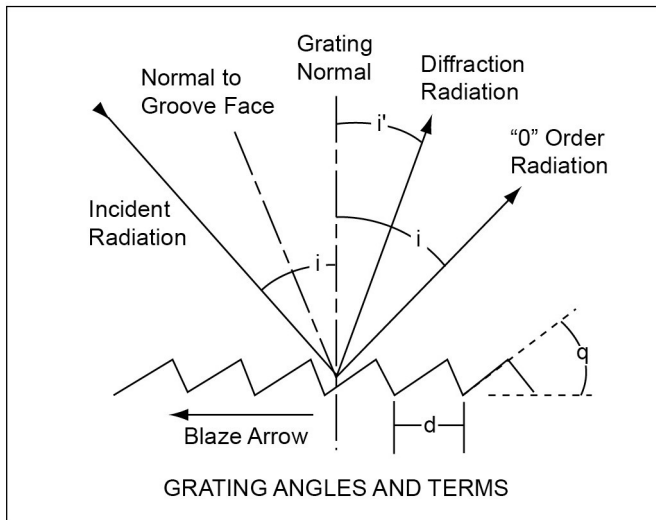
$$R = 15000$$



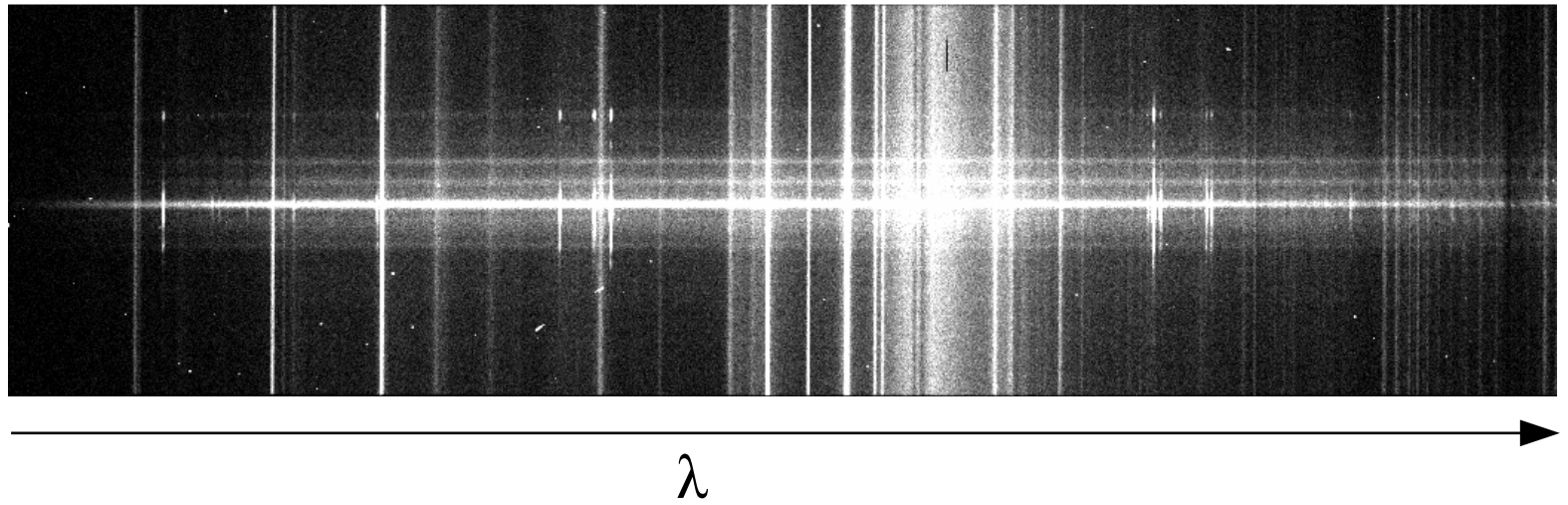
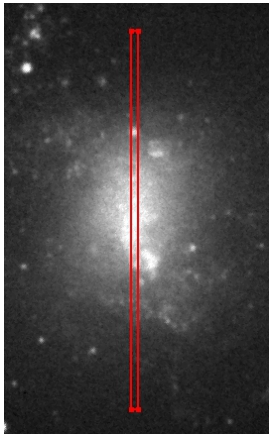
$$\frac{\Delta\lambda}{\Delta x} = 2.3 \text{ \AA/px}$$

$$R = 850$$

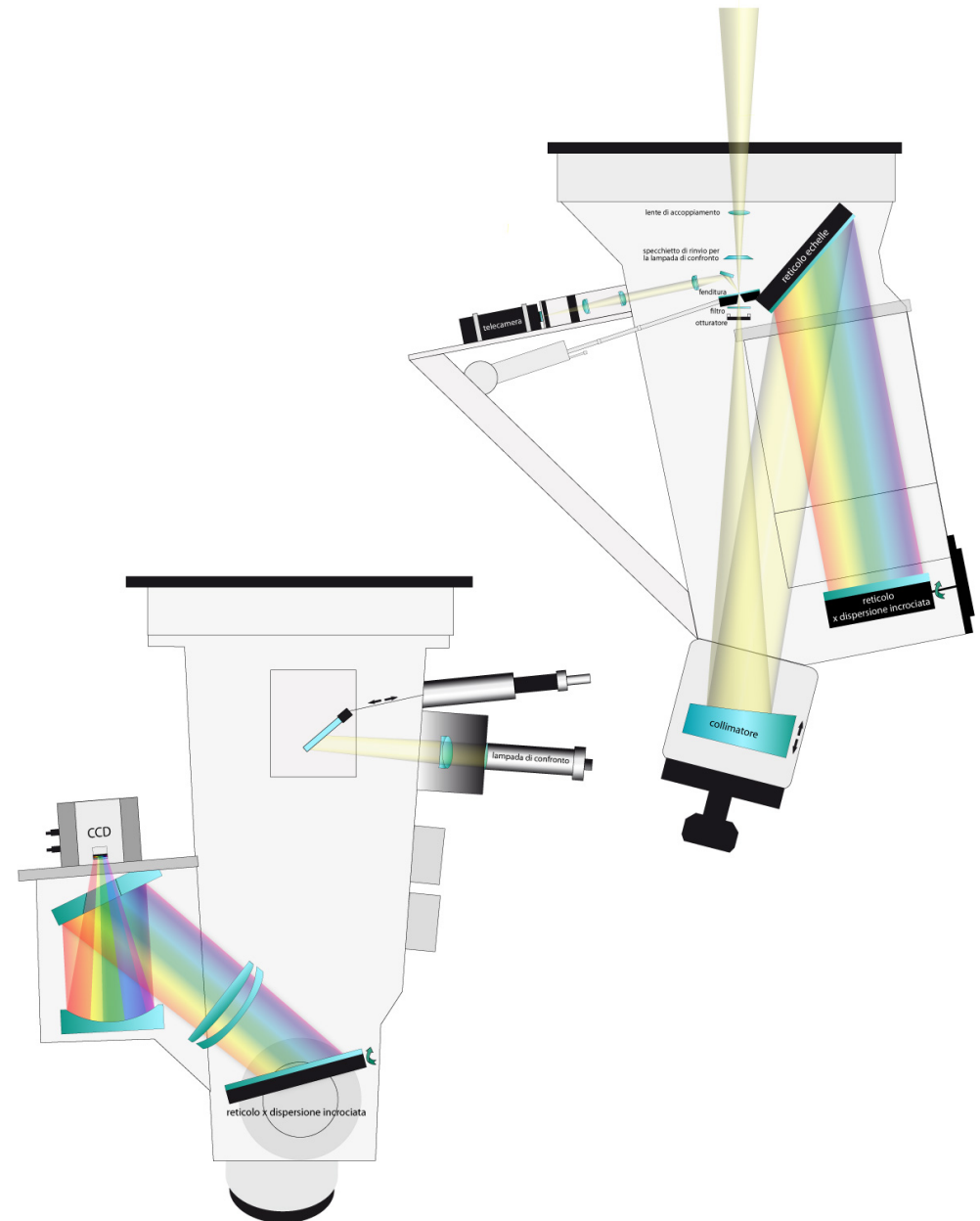
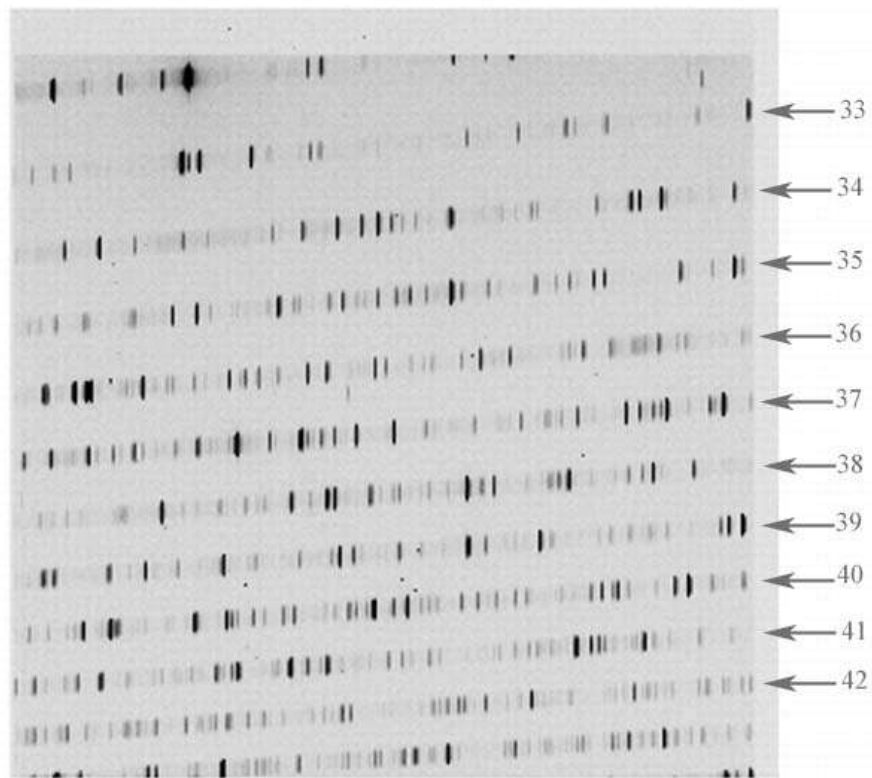
Grating blazed



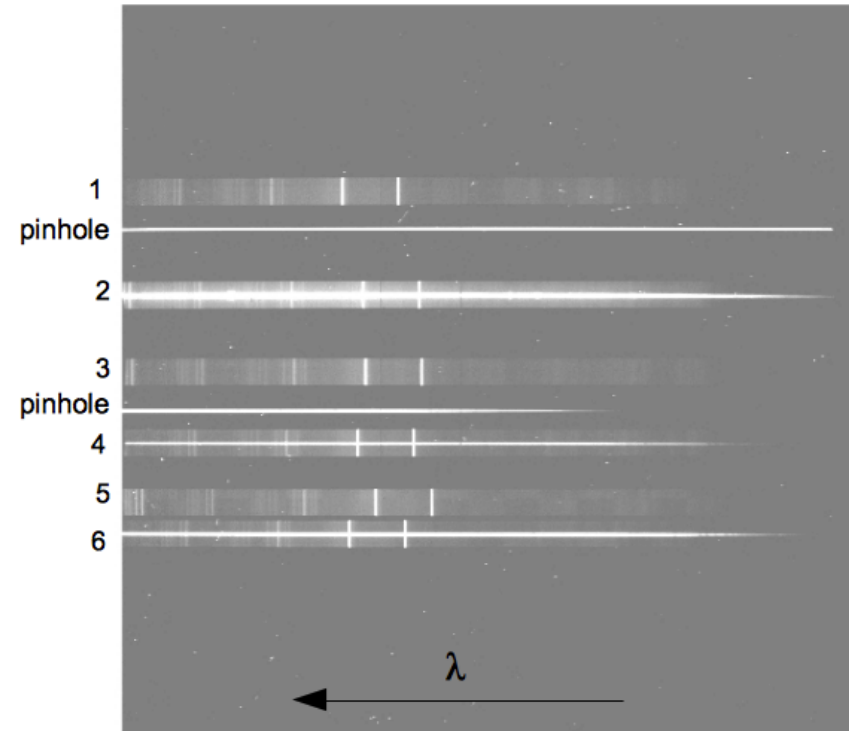
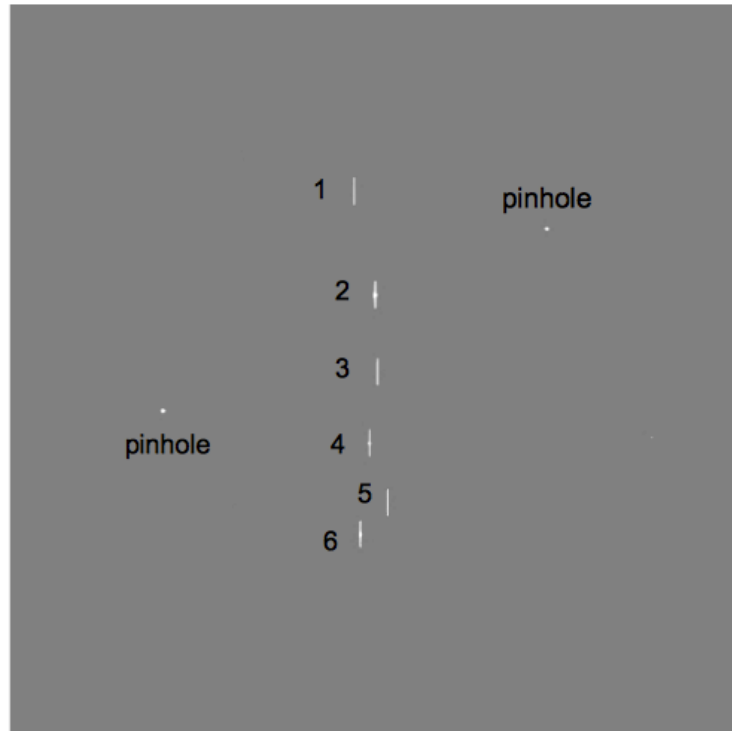
Longslit spectroscopy



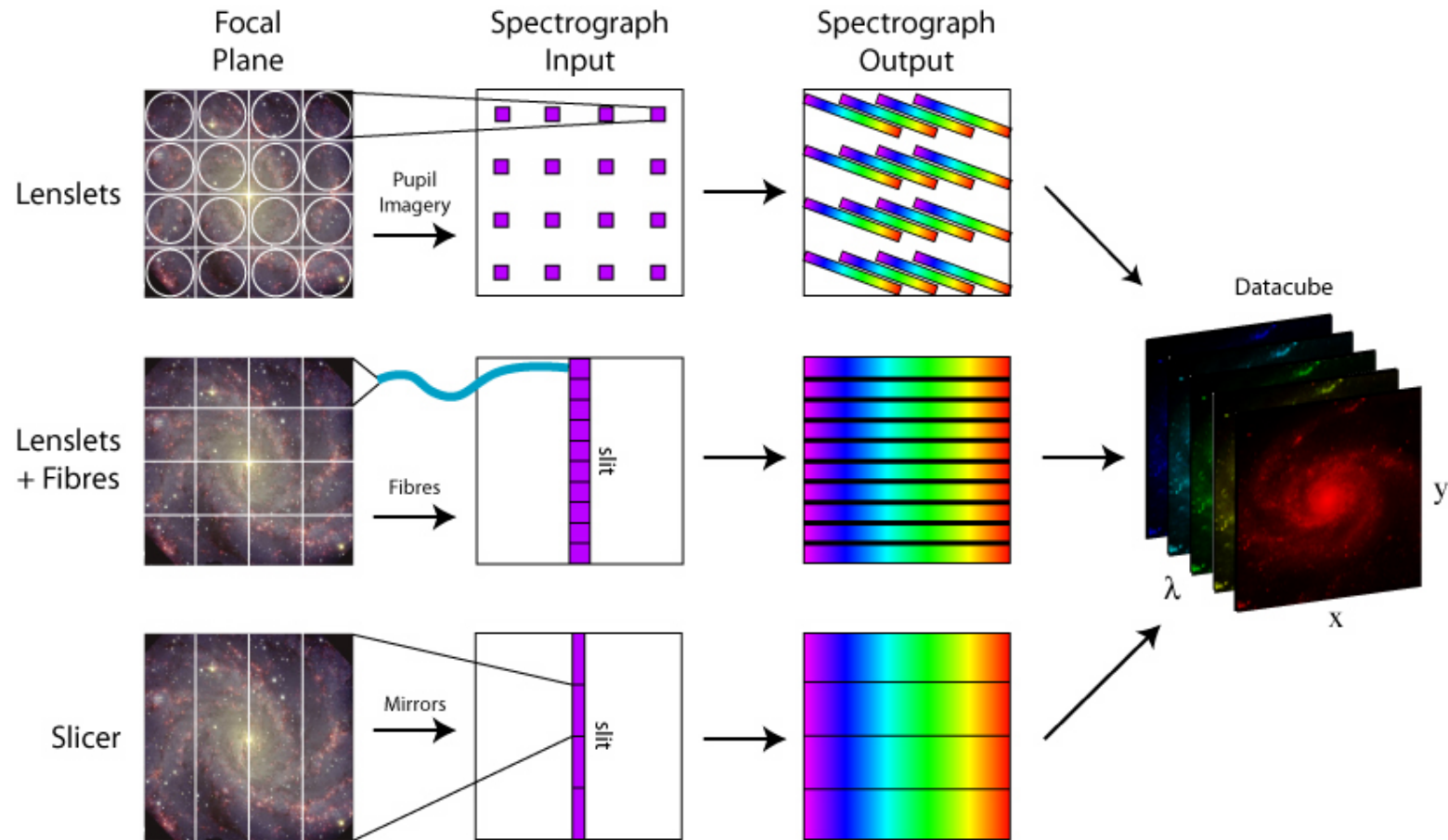
Echelle spectroscopy



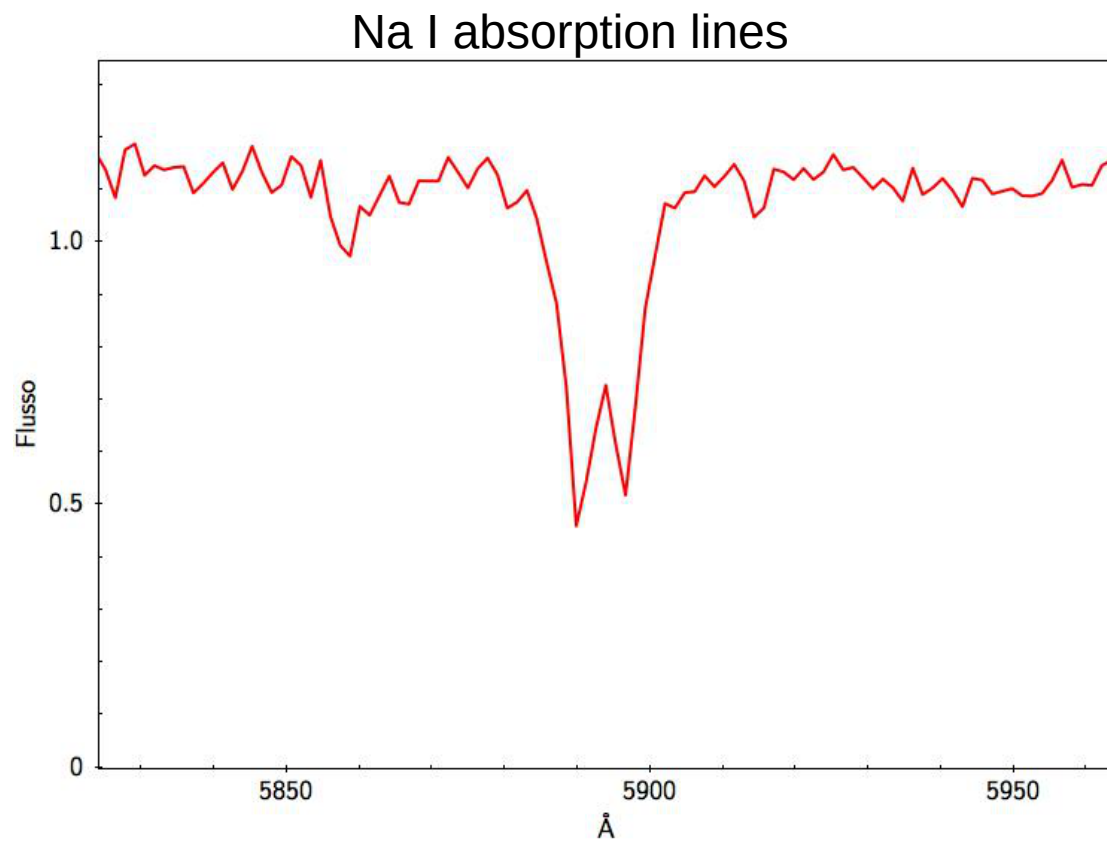
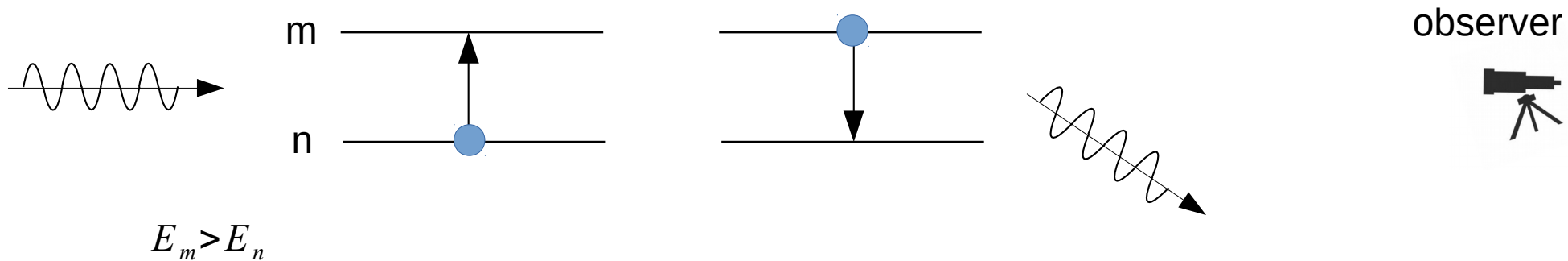
Multiobject spectroscopy

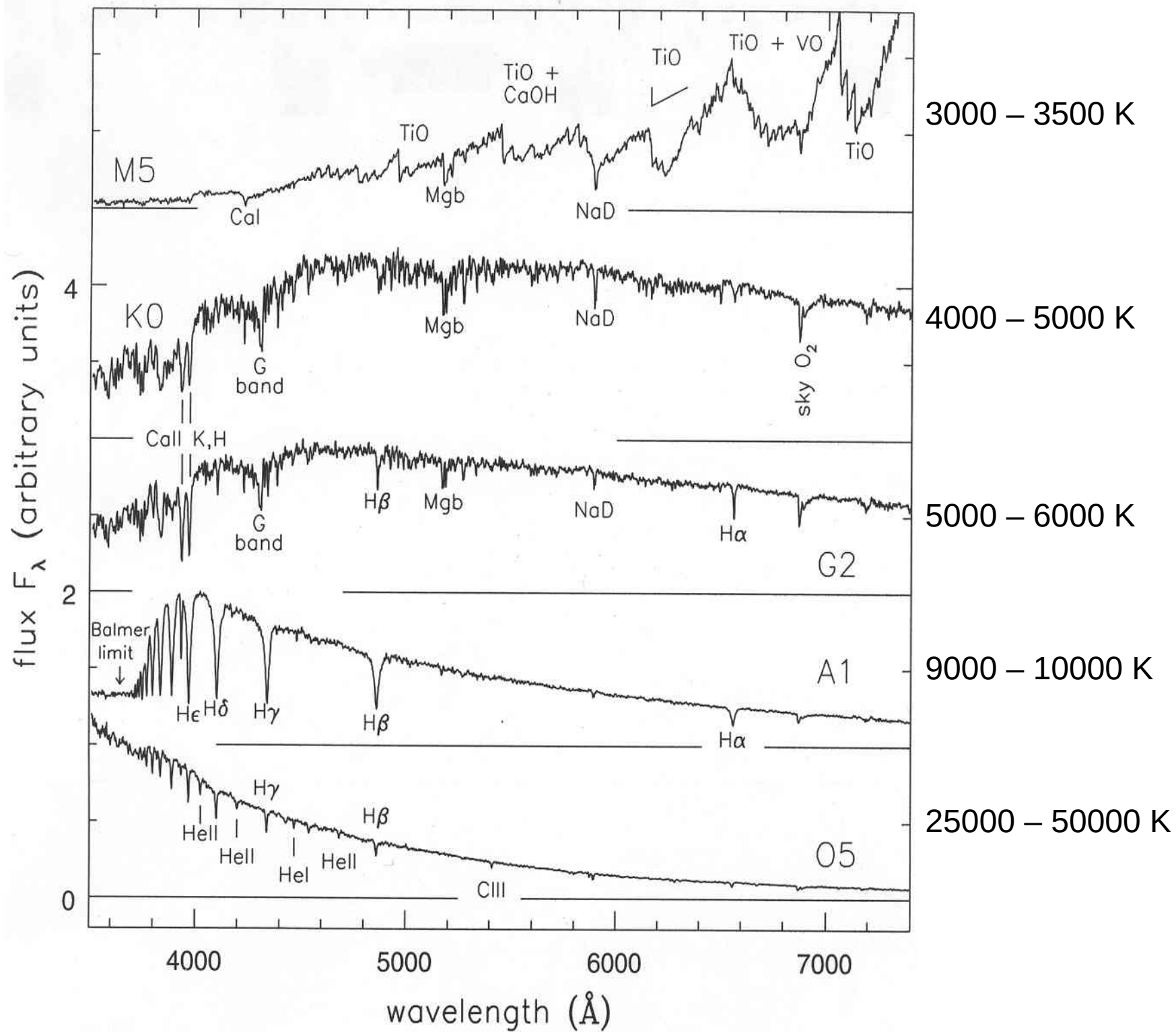


Integral-field spectroscopy



Stellar spectra





Stellar classification

Padre Angelo Secchi (1818-1878)



4000 stellar spectra

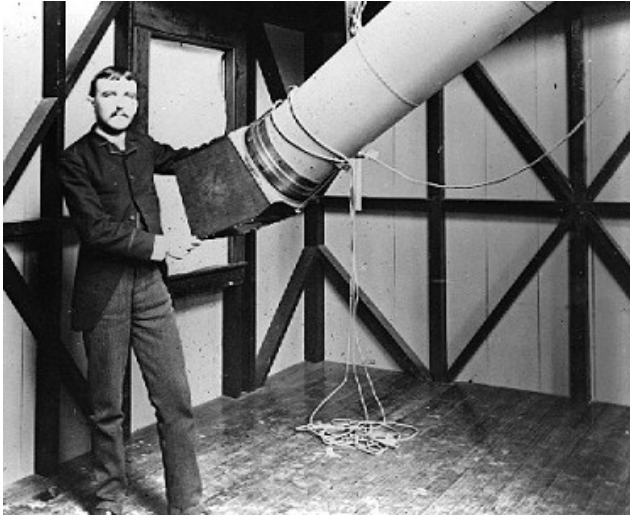


Secchi classes



Class I	White and blue stars with strong hydrogen lines
Class II	Yellow stars, hydrogen less strong, but evident metallic lines
Class III	Orange to red stars with complex band spectra
Class IV	Red stars with significant carbon bands and lines
Class V	Emission-line stars

Henry Draper (1837-1882)



Edward Pickering (1846-1919)



Harvard College Observatory (1899)

ANNALS

OF

THE ASTRONOMICAL OBSERVATORY OF HARVARD COLLEGE.

EDWARD C. PICKERING, DIRECTOR.

VOL. XXVII.

THE DRAPER CATALOGUE

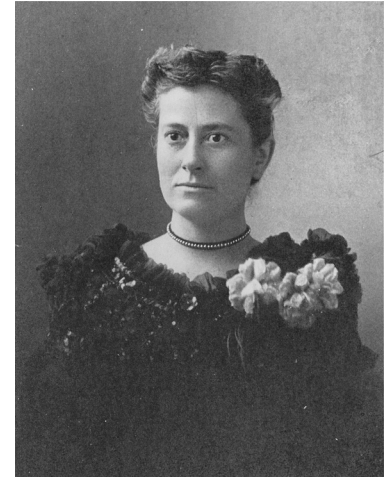
OF

STELLAR SPECTRA

PHOTOGRAPHED WITH THE 8-INCH BACHE TELESCOPE

The first of the photographs discussed in this volume were taken by Mr. H. Helm Clayton, the later ones by Mr. Willard P. Gerrish; those taken during the latter part of the night by Mr. R. W. Gifford. The reduction of the plates was begun by Miss N. A. Farrar, but the greater portion of this work, the measurement and classification of all the spectra, and the preparation of the Catalogue for publication, has been in charge of Mrs. M. Fleming. Misses F. Cushman, E. Gill, E. F. Leland, A. E. Masters, J. T. Rugg, M. C. Stevens, L. D. Wells and L. Winlock have at different times aided in the numerical computations and clerical work.

Williamina Fleming
(1857-1911)



1890 – Draper Catalogue of Stellar Spectra (10351 stars with $\delta > -25^\circ$)

Class I	A,B,C,D
Class II	E,F,G,H,I,L
Class III	M
Class IV	N
	O,P,Q

A N N A L S

OF

THE ASTRONOMICAL OBSERVATORY OF HARVARD COLLEGE

VOL. XXVIII.—PART II

SPECTRA OF BRIGHT SOUTHERN STARS

PHOTOGRAPHED WITH THE 13-INCH BOYDEN TELESCOPE

AS A PART OF

THE HENRY DRAPER MEMORIAL

AND DISCUSSED BY

ANNIE J. CANNON

UNDER THE DIRECTION OF

EDWARD C. PICKERING

DIRECTOR OF THE OBSERVATORY

CAMBRIDGE

Published by the Observatory

1901

Annie Jump Cannon
(1863-1941)



1122 stars

The letters of the Draper Catalogue are used in the following discussion to denote the various classes of spectra. The relation between the letters of the Draper Catalogue and the five types in ordinary use for visual spectra is as follows:—

Type.	Letter.	Type.	Letter.
I	A, B	III	M
I-II	F	IV	N
II	G	V	O
II-III	K		

hydrogen lines. The gradual decrease in the intensities of the Orion lines and the increase in the hydrogen lines were next made the basis of a series of subdivisions of Orion stars in which the interval between Class B and Class A is estimated in tenths. Thus the spectra of Class B 1 A are very nearly like those of Class B, while those called Class B 9 A show only slight differences from those of Class A. Those of Class B 5 A appear to be about midway between the two classes. The evidence that the Orion spectra precede the Sirian is as good as that the Sirian precede the solar. The gradual decrease in the intensities of the Orion lines is accompanied by gradual increase in the hydrogen lines, and by the incoming of faint solar lines, so that in spectra of Classes B 8 A and B 9 A, solar and Orion lines are commingled. Hence, it was necessary either to interchange the letters B and A of the Draper Catalogue or to place the letter B before the letter A. The first alternative would prove confusing. The second presents no real difficulties since the letters are merely symbols to express an observed condition.

preceding those of the Orion type. The letter O has been placed, therefore, in this classification, before the letter B instead of after the letter M. The letter P represents the spectra of planetary nebulae. The letter Q represents peculiar spectra having bright lines.

No. 1

STELLAR ATMOSPHERES

A CONTRIBUTION TO THE OBSERVATIONAL
STUDY OF HIGH TEMPERATURE IN THE
REVERSING LAYERS OF STARS

BY

CECILIA H. PAYNE

PUBLISHED BY THE OBSERVATORY

CAMBRIDGE, MASSACHUSETTS

1925



Cecilia Payne-Gaposchkin(1900-1979)

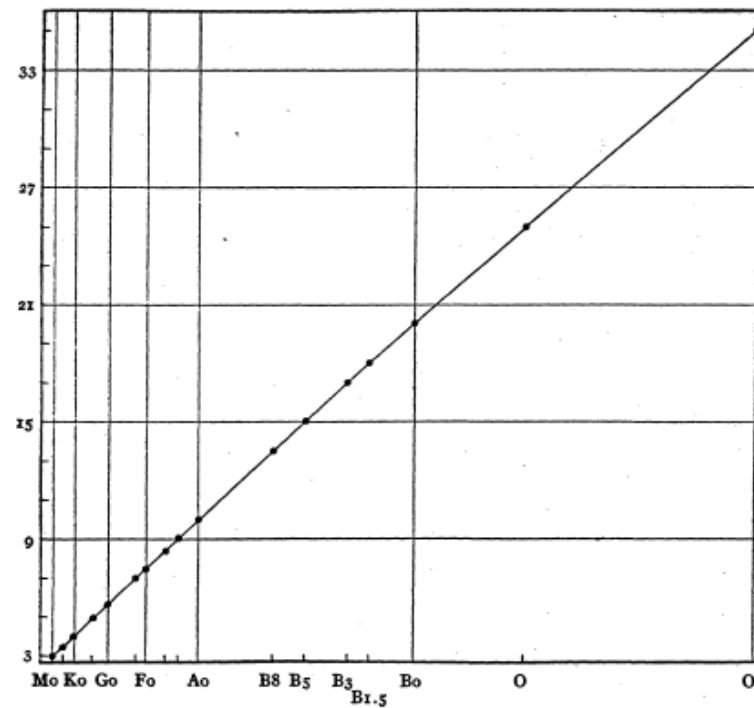


FIGURE 10

Schematic representation of the ionization temperature scale for the sequence of stellar classes. Ordinates are absolute temperatures in thousands of degrees; abscissae are Draper classes. The points representing the different classes have been made to lie on a straight line, so that the temperature range of the corresponding classes shall appear along the axis of abscissae. Vertical lines are drawn through Mo, Ko, Go, Fo, Ao, Bo, and the upper limit of the O class, in order to show the increase in temperature range for the hotter classes.



William Morgan
(1906-1994)



Philip Keenan
(1908-2000)

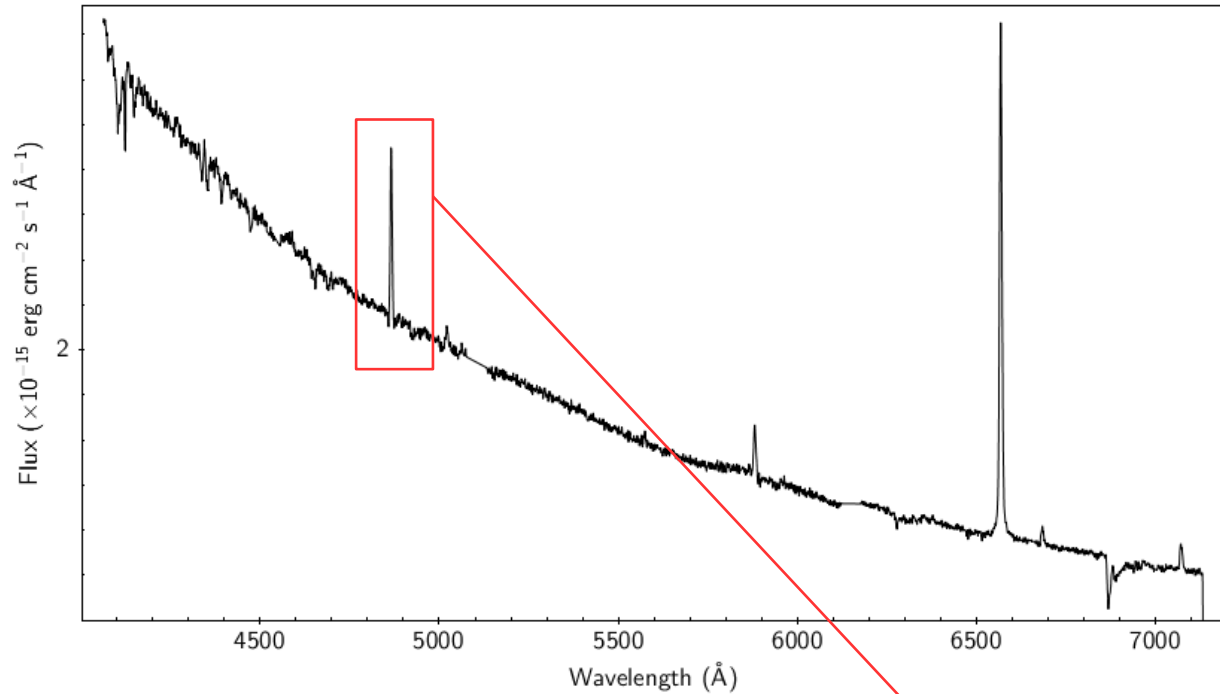
Edith Kellman
(1911-2007)

Yerkes Observatory staff (1938)

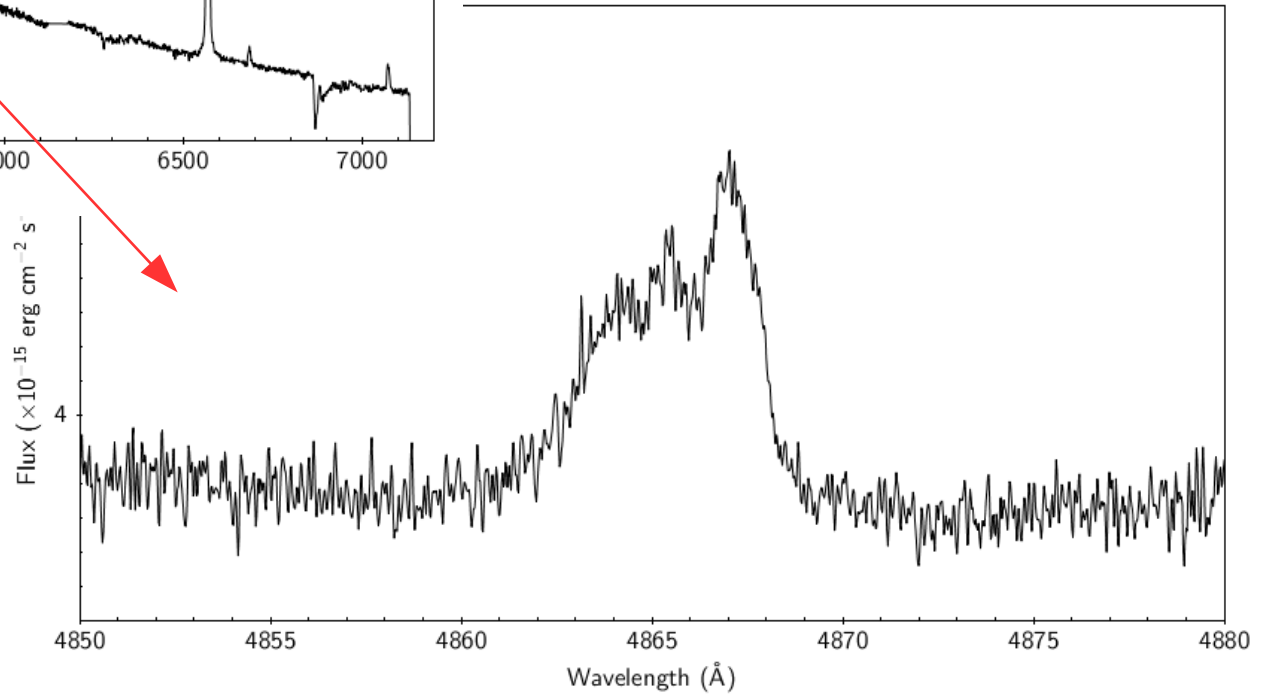
- 1943 – MKK system
- 1953 – Morgan – Keenan system

Luminosity classes: the combined effects of temperature and gravity on spectral lines

Emission Line Stars

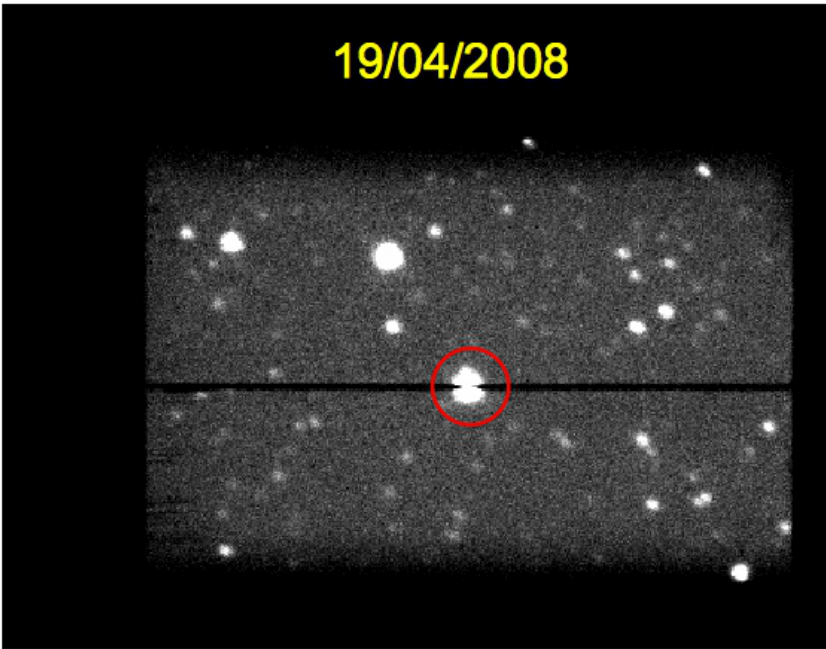


Example of Be star

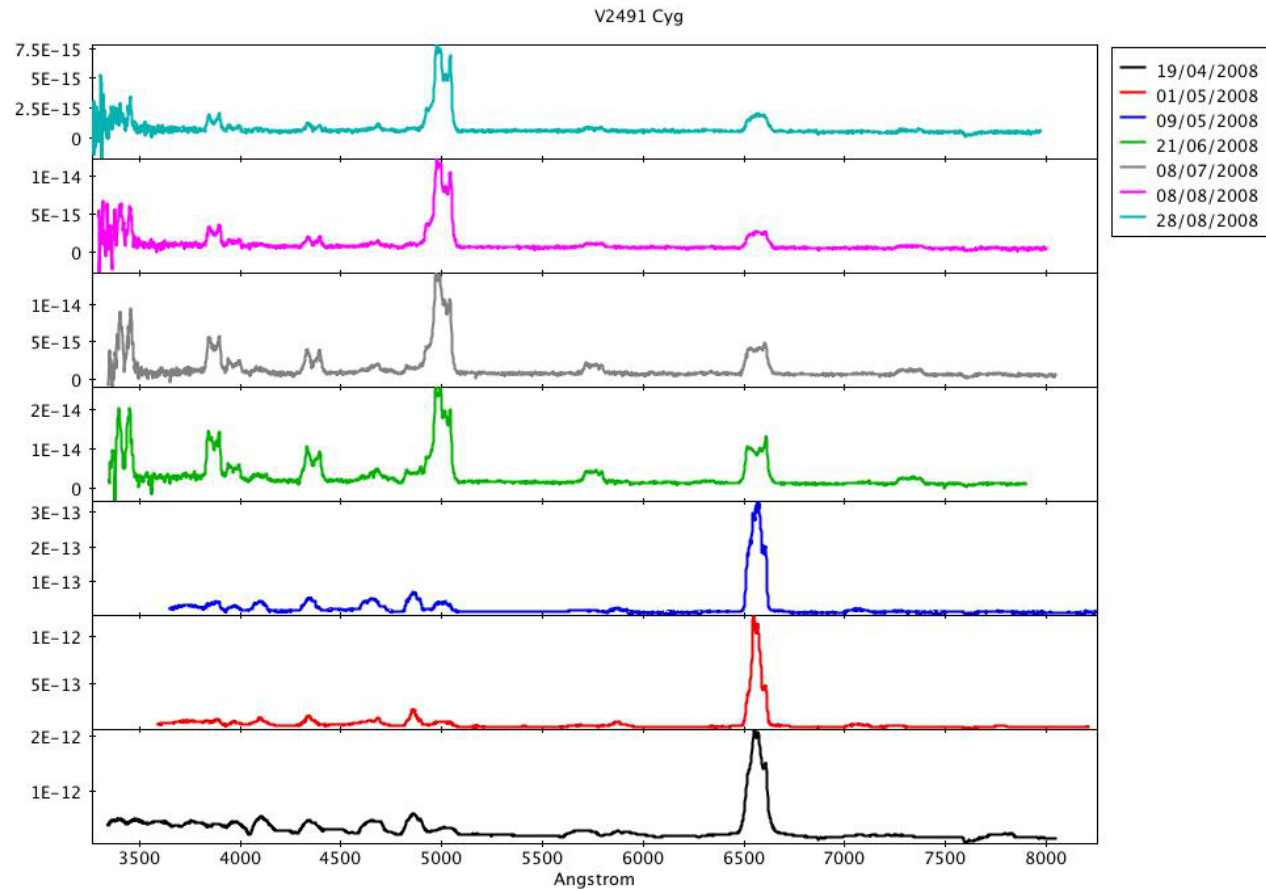
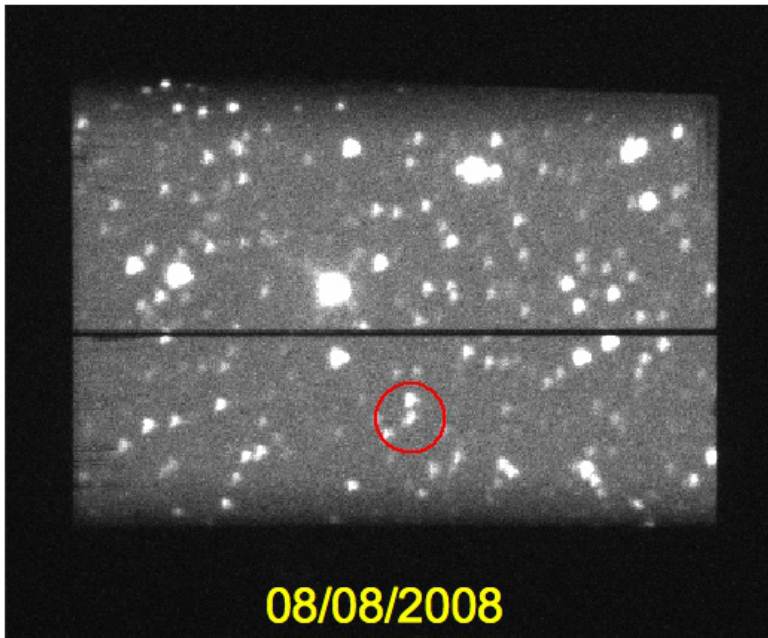


Example of Galactic nova: V2491 Cyg

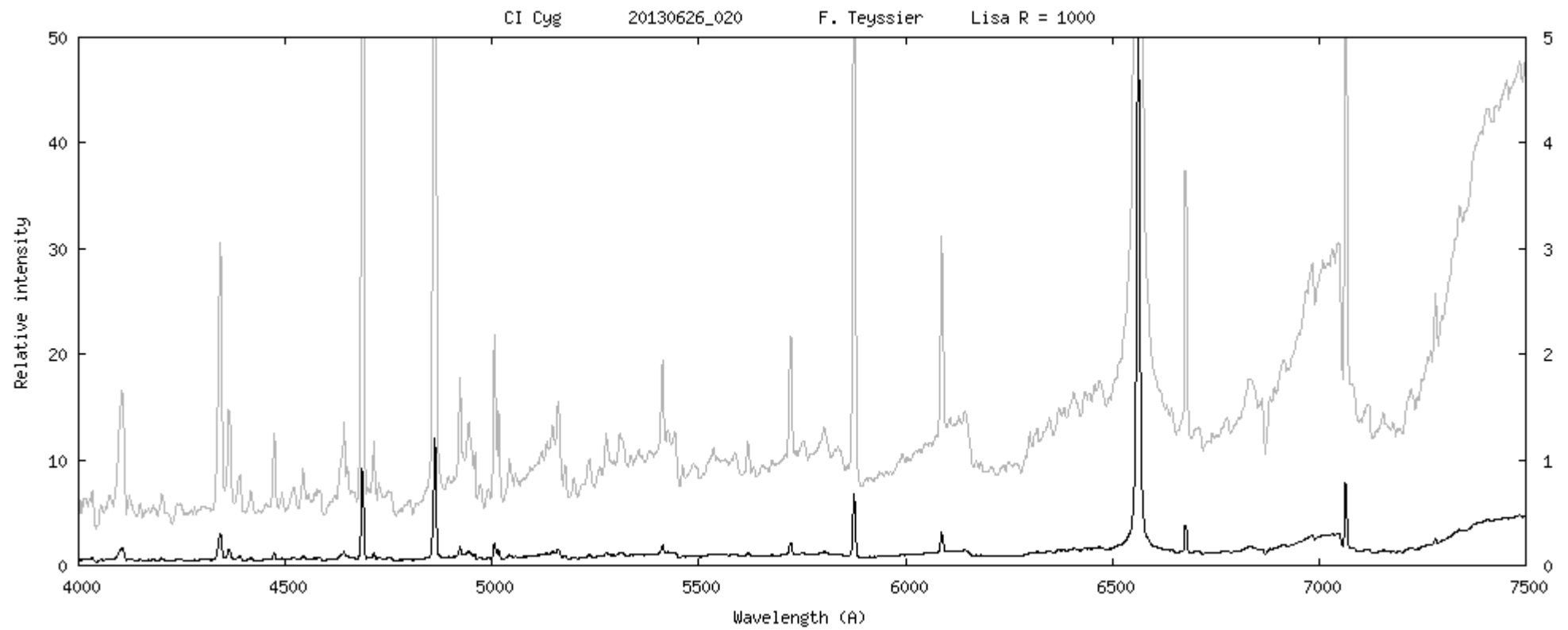
19/04/2008



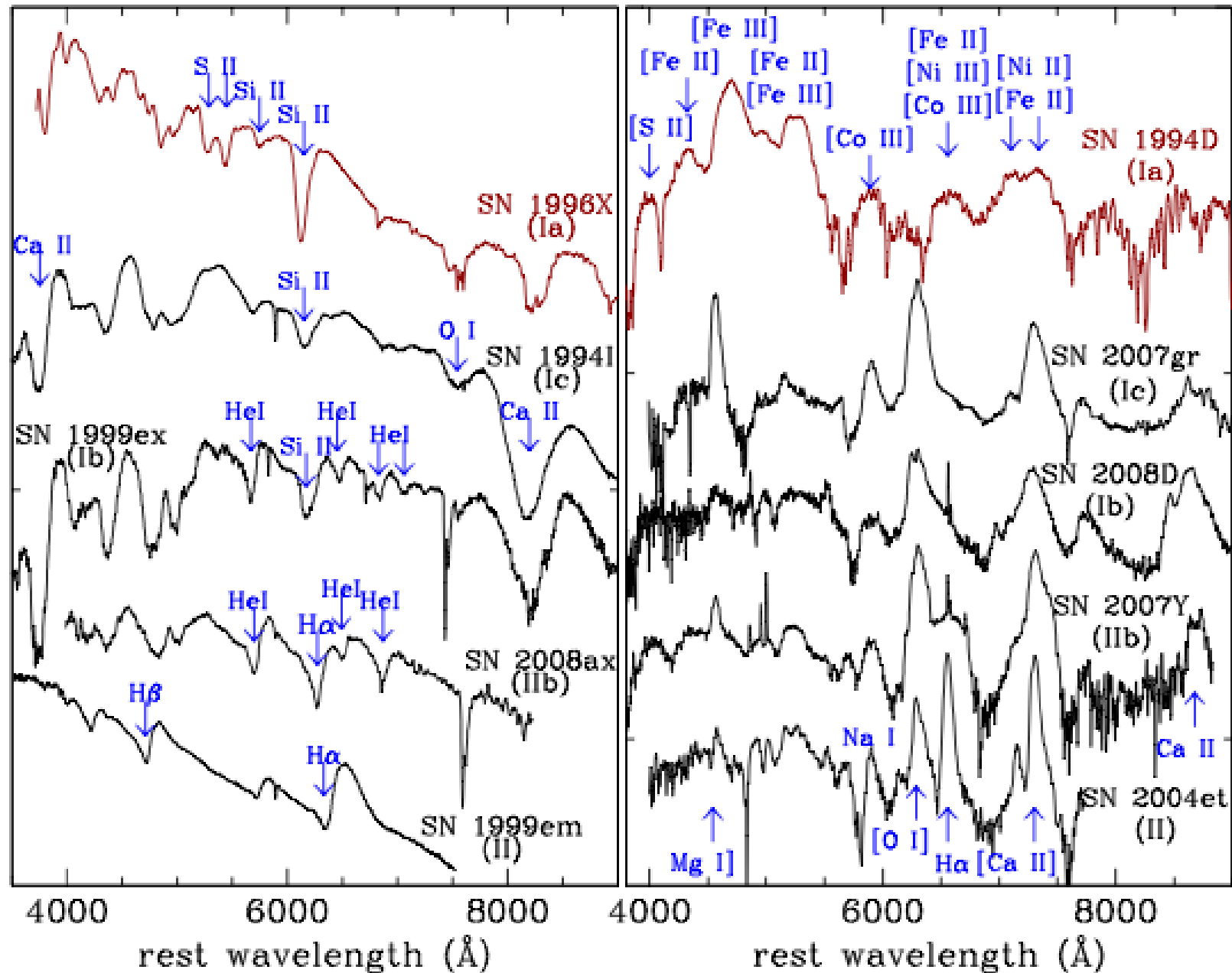
08/08/2008



Example of symbiotic star: CI Cyg

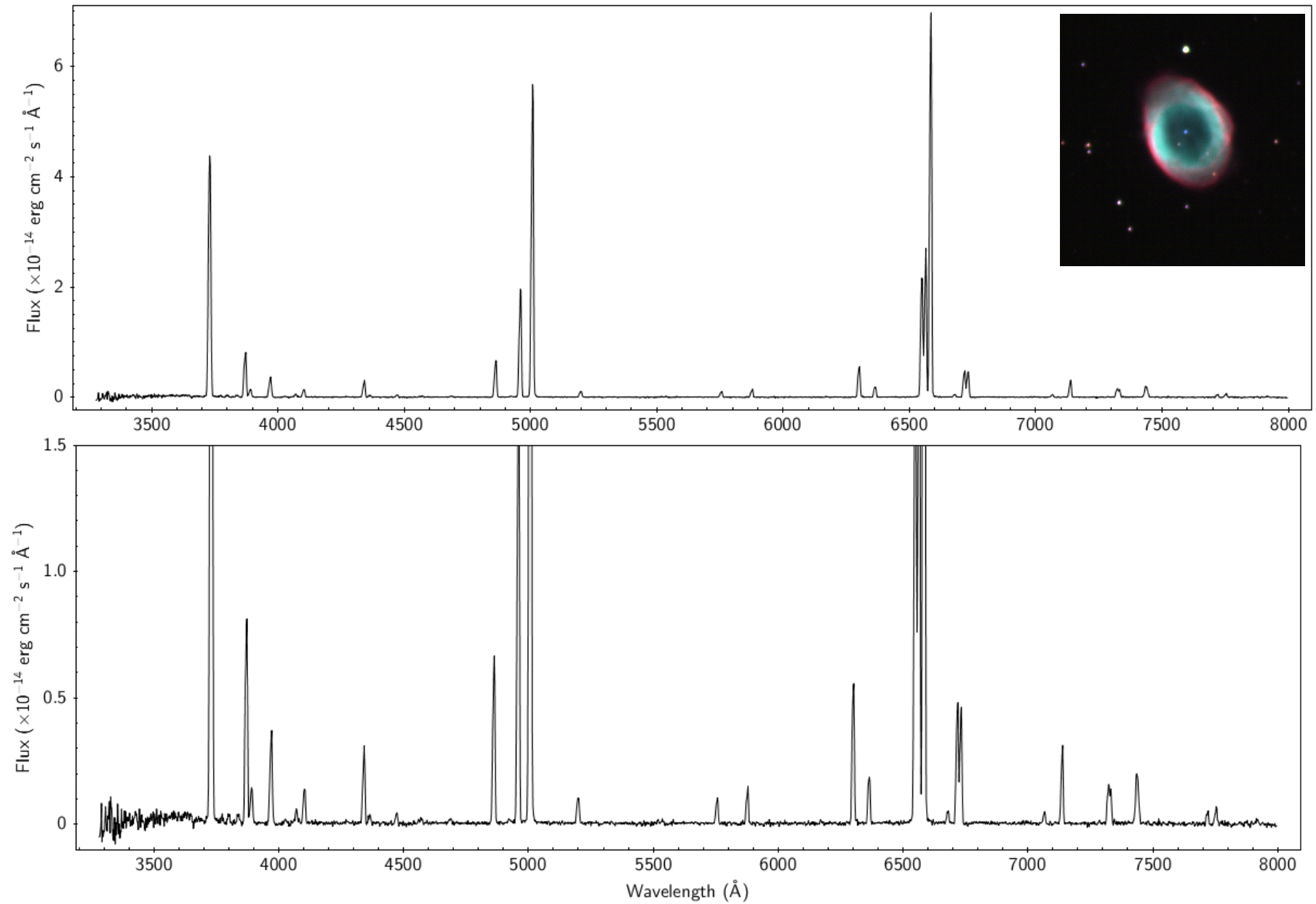


Examples of supernovae spectra

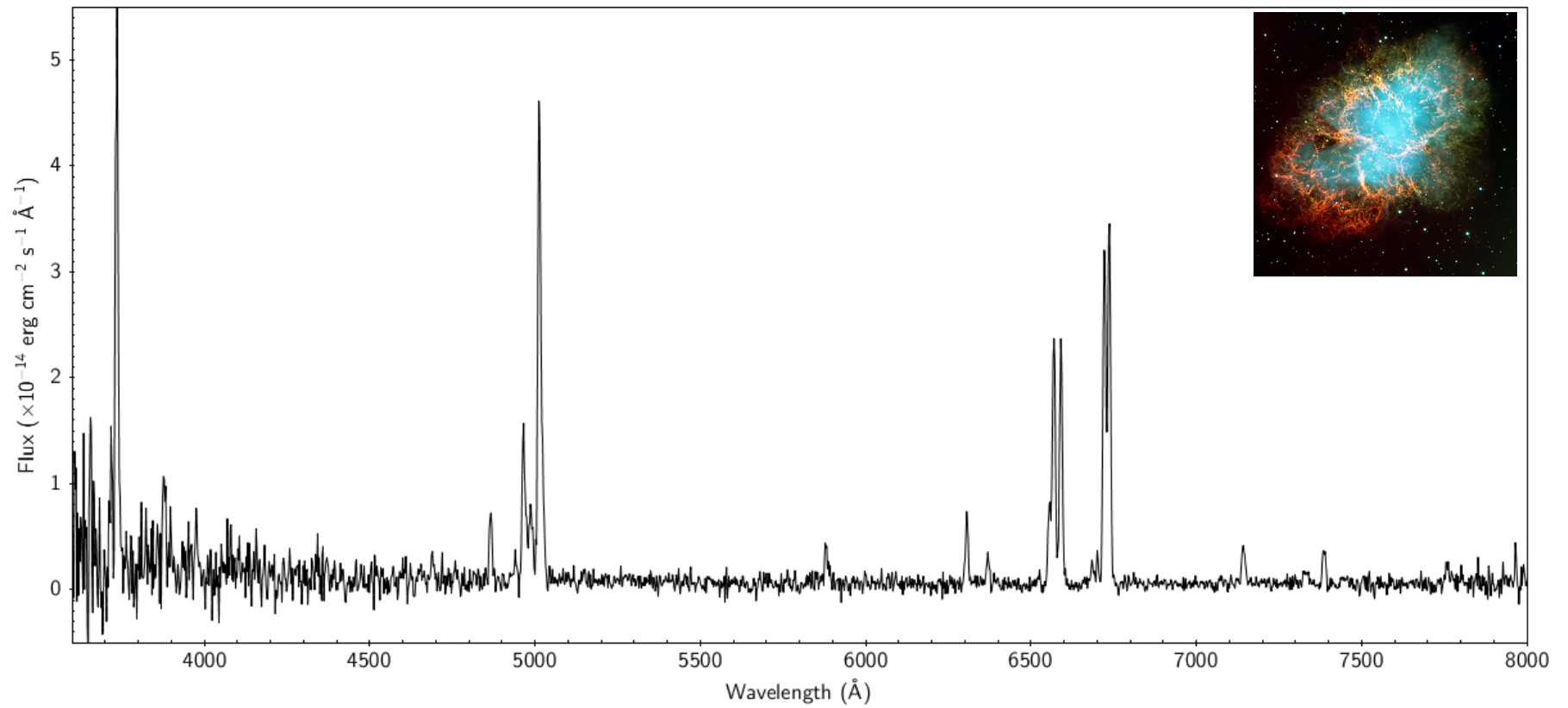


H II regions

Example of planetary nebula: M57

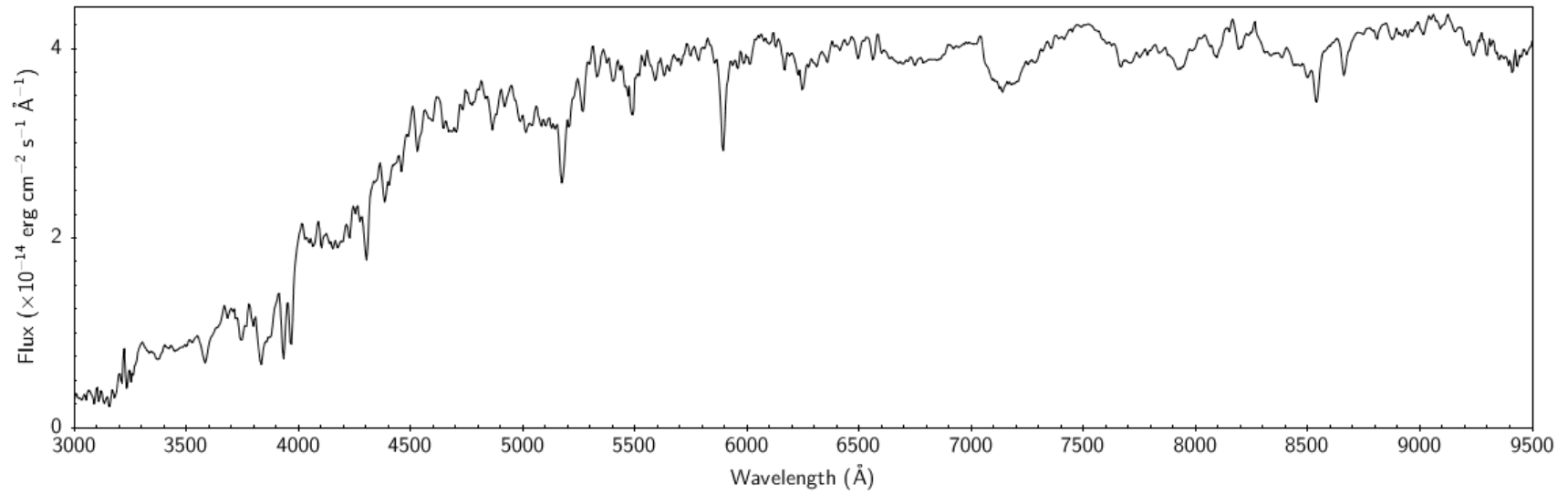


Example of supernova remnant: M1

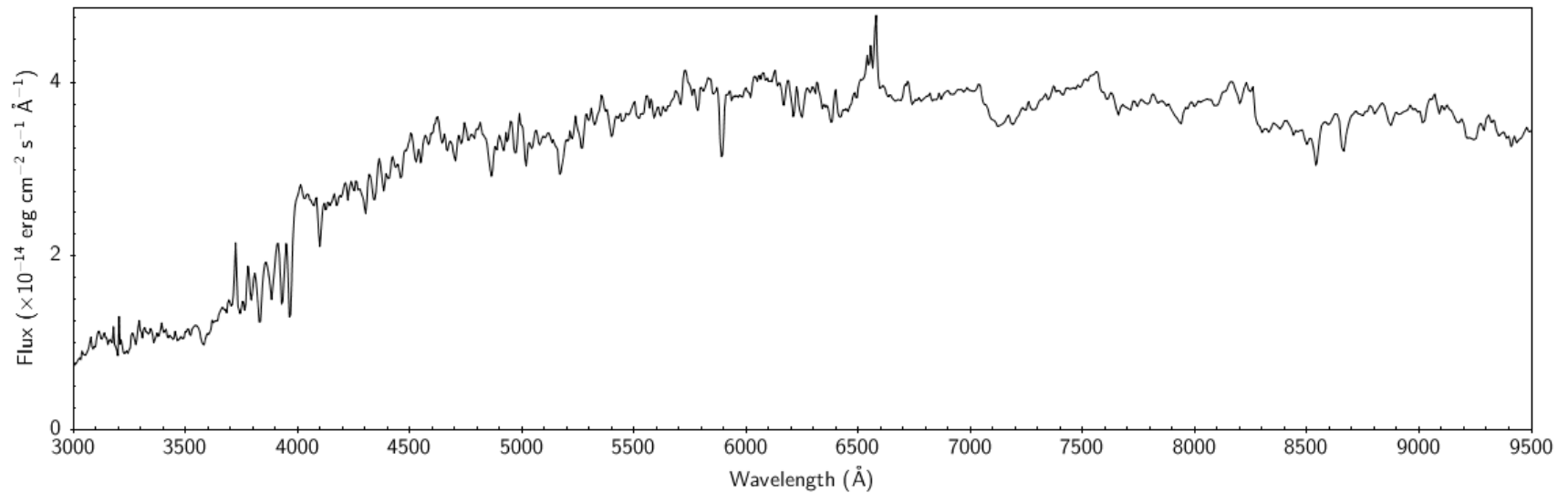


Galaxies

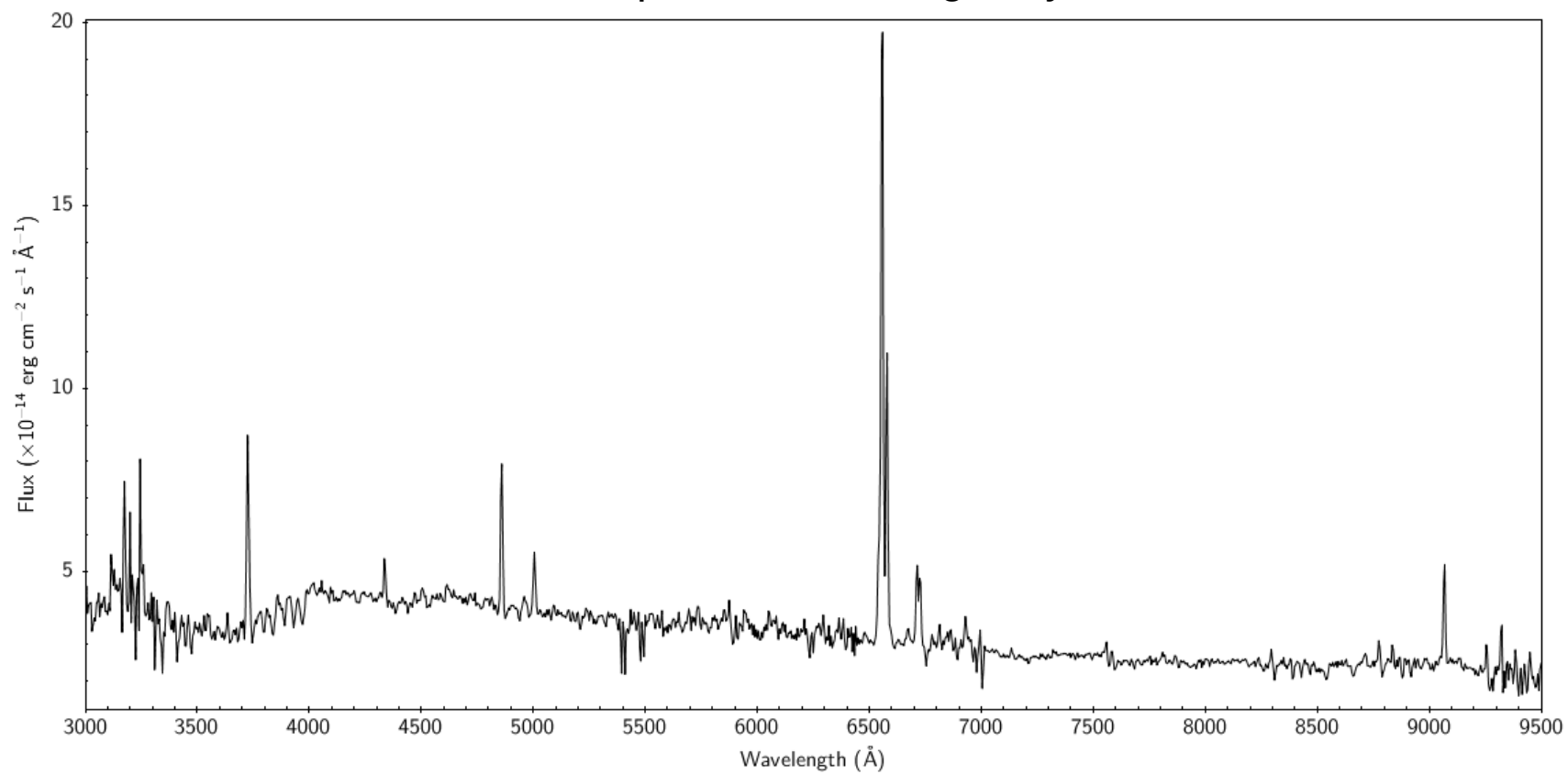
Example of elliptical galaxy



Example of Sa galaxy



Example of starburst galaxy



Examples of AGN

